Astron. Astrophys. 214, 186–202 (1989)

ASTRONOMY AND ASTROPHYSICS

Evolution of close binary systems that undergo a dynamically stable late case C mass transfer

L. Pastetter and H. Ritter

Universitäts-Sternwarte München, Scheinerstrasse 1, D-8000 München 80, Federal Republic of Germany

Received August 29, accepted September 29, 1988

Summary. It is shown that for a restricted range of initial binary parameters the originally more massive primary can reduce its mass via a stellar wind on the AGB so much, i.e. to less than half of its initial mass, that at the onset of Roche-lobe overflow the binary system will be dynamically stable against mass transfer. Numerical simulations of systems during the corresponding late case C mass transfer yield the following results: Computations done with a standard technique for computing the mass transfer give incorrect results. The failure of the standard method is traced back to the fact that the algorithm used assumes implicitly a sharp stellar rim, whereas the atmospheric scale height of the very extended primary ($R \approx 500\,R_\odot$) is $H \approx 0.05R \approx 25\,R_\odot$. Computations done using an improved technique that takes the finite atmospheric scale height explicitly into account yield results that are qualitatively different from those obtained by using the standard method. While the latter yields spasmodic mass transfer at each thermal pulse (with no mass transfer in between) the former yields sustained mass transfer, modulated in time by the thermal pulses. Mass transfer rates of the order of $10^{-5} \ldots 10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$ and a wind loss rate of $\approx 10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$ are obtained. At these mass transfer rates the primary underfills its critical Roche volume significantly. The duration of the mass transfer phase is of the order 10^5 yr. The accretion disk that forms around the companion star is very extended ($R_\mathrm{D} \approx 500 - 1000\,R_\odot$), its outer parts are very cool ($T_\mathrm{D} < 10^3$ K) and optically thick. The main source of the opacity is probably dust. The total accretion luminosity of the disk is always very small compared to the primary's luminosity. Therefore, such a disk remains essentially invisible and can manifest itself only by eclipsing the primary in systems of sufficiently high orbital inclination.

Key words: close binaries – stellar evolution – mass transfer – accretion disks

1. Introduction

Since the first detailed numerical studies of the evolution of close binaries by Paczyński (1966) and Kippenhahn and Weigert (1967) an almost uncountable number of papers have been devoted to binary evolution (for reviews see e.g. Thomas, 1977; Webbink, 1979b, 1985). With very few exceptions these studies have dealt with either case A or case B mass transfer (in the terminology introduced by Kippenhahn and Weigert, 1967). On the other hand, a third type of binary evolution, hereafter case C (following Lauterborn, 1970), has found comparatively little attention so far. To understand the reasons for this it is necessary to characterize a case C evolution in some more detail. According to Lauterborn (1970), a case C evolution involves the onset of Roche lobe overflow (hereafter RLOF) from the (more massive) primary after the end of its central helium burning but before ignition of carbon burning. If the primary was a star of low or intermediate mass on the main sequence, i.e. $M_1 \lesssim 8\text{–}10\,M_\odot$, case C implies that at the onset of RLOF the primary is on the asymptotic giant branch (AGB), where it has a deep outer convection zone that extends in mass almost to the degenerate carbon oxygen core. In the standard theory of binary evolution the primary evolves at constant mass until the onset of mass transfer. Therefore, the primary is the more massive component, not only on the main sequence, but still at the beginning of ROLF. The time scale of the subsequent mass transfer can be anything between the nuclear and the dynamical time scale of the primary, depending on the mass ratio M_1/M_2 and on the response of the primary's radius to mass loss. Now, in a case C evolution, the primary has a deep outer convective envelope. As a result, the stellar radius has a tendency to increase in response to mass loss, particularly to rapid mass loss. Because of this and because $M_1 > M_2$ at the onset of ROLF, it was soon realized that case C mass transfer would unavoidably lead to mass transfer on a dynamical (or at least convective) time scale. Since mass transfer at such extremely high rates could not (and still cannot) be dealt with adequately in numerical simulations, case C, being intractable, was put aside. There are, however, a few exceptions as we mentioned above.

The first attempt to model a case C evolution was made by Lauterborn (1970). Not surprisingly he could not follow the phase of rapid mass transfer adequately. In order to proceed he bridged the fast phase by ad hoc mass loss of the primary. For the subsequent slow phase of mass transfer (on a nuclear time scale) he assumed the primary to remain in thermal equilibrium. In this way, the thermal pulses and their possible influence on the mass transfer were suppressed. That even the onset of rapid mass transfer in a case C evolution can be awfully complicated has been shown by Webbink (1979a). Later, Iben (1986) performed extensive numerical computations of early case C evolutions where mass transfer starts before the onset of the thermal pulses. Here again the fast phase of mass transfer was not followed appropriately. Finally de Kool et al. (1986) studied case C mass transfer by extending work of Webbink et al. (1983) on low-mass

Send offprint requests to: H. Ritter

case B mass transfer towards higher core masses. In their study, de Kool et al. (1986) circumvent the problems related to the dynamical mass transfer instability and to the thermal pulses by assuming binary parameters that allow for a stable mass transfer (without discussing which type of evolution would result in such parameters), and by using a core-mass luminosity and a core-mass radius relation, rather than full stellar models, to model the evolution of the primary.

The starting point of this paper is the observation that stars with an initial mass $M_i \lesssim (8 \pm 2)\,M_\odot$ on the main sequence become white dwarfs with a mass $M_f < 1.4\,M_\odot$ (e.g. Weidemann and Koester, 1983). In such an evolution the bulk of the mass is lost via a stellar wind on the AGB. In the context of binary evolution this means that it is not necessarily true that the primary evolves with constant mass until the onset of RLOF, particularly not for case C where this happens when the star is already on the AGB. Therefore, we may speculate whether the originally more massive primary could reduce its mass via a stellar wind on the AGB so much that at the onset of RLOF it is not only the less massive component but that mass transfer will also be dynamically stable. This question will be clarified in Sect. 2 where we also determine quantitatively the prerequisites for a dynamically stable case C mass transfer. In Sect. 3 we present results from numerical computations, using full stellar models and standard techniques to compute the mass transfer. In these computations the effects of thermal pulses are also taken into account. The results we obtain in Sect. 3 indicate that the standard technique for computing mass transfer is inadequate for the stars in question. Therefore we introduce an improved scheme for computing mass transfer in Sect. 4. In Sect. 5 we present results of numerical simulations of RLOF using the improved technique. In most cases, mass transfer in a close binary gives rise to the formation of an accretion disk around the companion star. The properties of this disk in systems with case C mass transfer and its possible feed back on the mass transfer via spin orbit coupling are discussed in Sect. 6. Based on the results of our computations we try to sketch a rough picture of binary systems in a phase of a dynamically stable late case C mass transfer in the discussion (Sect. 7). The possible consequences on our results of several approximations and simplifications made in this paper are also discussed in Sect. 7. Finally the conclusions of this paper are presented in Sect. 8.

2. Prerequisites for a dynamically stable case C evolution

2.1. Stability against dynamical time scale mass transfer

We consider a semi-detached binary system with components of mass M_1 and M_2, respectively, in which the primary fills its critical Roche lobe. The criterion for stability against runaway mass-transfer on a dynamical time scale, hereafter referred to as dynamical stability, is (e.g. Webbink, 1985)

$$\zeta_{1,S} = \left(\frac{\partial \log R_1}{\partial \log M_1}\right)_S > \left(\frac{\partial \log R_{1,R}}{\partial \log M_1}\right) = \zeta_{R,1}\,. \tag{1}$$

Here $\zeta_{1,S}$ is the adiabatic mass radius exponent of the primary. The subscript S indicates that the response of the stellar radius to mass loss has to be evaluated keeping the Lagrangian entropy profile $S(M_r)$ fixed throughout the star. In order to evaluate $\zeta_{R,1}$, i.e. the response of the primary's Roche-lobe radius $R_{1,R}$ to mass loss, one must first specify how mass and angular momentum of the binary is lost from or redistributed in the system as a consequence of mass transfer and other evolutionary effects.

Here we envisage the following situation: The primary, an AGB star, may loss mass via Roche-lobe overflow at a rate $\dot M_{1,\mathrm{tr}}$ and at the same time via a stellar wind at a rate $\dot M_{1,\mathrm{w}}$. We assume the stellar wind to emanate isotropically from the primary and to carry with it the specific angular momentum at the primary's center of gravity. As a result, the system loses orbital angular momentum J_{orb} at a rate

$$(\dot J_{\mathrm{orb}})_{\mathrm{w}} = \omega A_1^2 \dot M_{1,\mathrm{w}}\,, \tag{2}$$

where ω is the orbital angular frequency and A_1 the distance of the centre of gravity of the primary to that of the binary. In addition to this, mass transfer can extract angular momentum from the orbit and store it in an accretion disk, at least temporarily. The corresponding rate of loss of angular momentum is

$$(\dot J_{\mathrm{orb}})_{\mathrm{tr}} = f X^2_{\mathrm{L}_1,M_2} A^2 \omega \dot M_{1,\mathrm{tr}}\,, \tag{3}$$

where X_{L_1,M_2} is the distance of the inner Lagrangian point L_1 from the secondary in units of the orbital separation A and f is a factor of order unity that takes into account the fact that matter leaving L_1 towards the secondary does not conserve exactly its angular momentum with respect to M_2 (e.g. Lubov and Shu, 1975; Flannery, 1975; where f is tabulated as a function of the mass ratio).

Introducing the abbreviation

$$k = \frac{\dot M_{1,\mathrm{w}}}{\dot M_{1,\mathrm{tr}}} \tag{4}$$

and the mass ratio

$$q = \frac{M_1}{M_2} \tag{5}$$

we find

$$\zeta_{R,1} = \frac{2k}{(k+1)(q+1)} + \frac{2f(1+q)X^2_{\mathrm{L}_1,M_2}}{k+1} + \left(\frac{q}{k+1}+1\right)\beta_1 + \frac{q(2q+1)}{(k+1)(q+1)} - \frac{q+2}{q+1}, \tag{6}$$

where

$$\beta_1 = \frac{d\ln f_1(q)}{d\ln q}. \tag{7}$$

$$f_1(q) = \frac{R_{1,\mathrm{R}}}{A} \tag{8}$$

is the primary's Roche radius in units of the orbital separation and depends only on the binary's mass ratio. Numerical values of $f_1(q)$ may be obtained either from the tables of Plavec and Kratochvil (1964) or Mochnacki (1984) or from analytical approximations (e.g. Eggleton, 1983). Figure 1 shows $\zeta_{R,1}$ as a function of q for various values of the parameter k with $f = 0$.

Whether or not the primary is stable against mass transfer depends now on the mass radius exponent ζ_1 of the lobe-filling primary. Therefore, we have to examine ζ_1 for the type of primary in question, i.e. for a star on the AGB.

Let us first examine the response of the stellar radius to mass loss that is sufficiently slow, allowing the star to remain in ther-

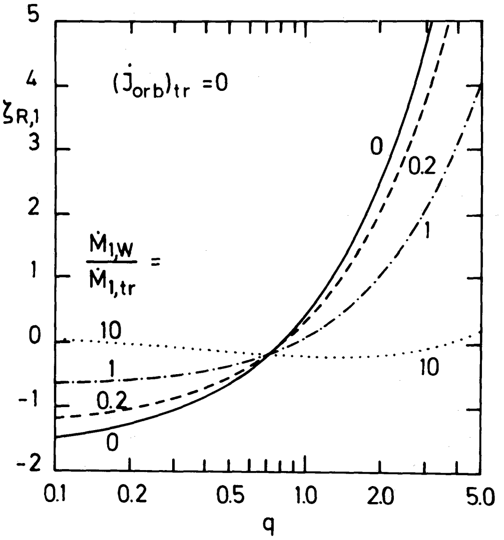

Fig. 1. $\zeta_{R,1}$ as a function of the mass ratio $q = M_1/M_2$ according to Eq. (6) with $f = 0$, i.e. no storage of the transferred angular momentum in the accretion disk, for different values of the ratio $\dot{M}_{1,w}/\dot{M}_{1,tr}$. Note that an adiabatic mass radius exponent of the primary of $\zeta_{1,s} \approx -\frac{1}{3}$ results in a critical mass ratio $q_{crit} \approx 0.65$ with only a weak dependence of q_{crit} on $\dot{M}_{1,w}/\dot{M}_{1,tr}$ as long as this ratio is smaller than about unity

mal equilibrium. It is well known that double shell source burning equilibrium models follow a core-mass luminosity relation (e.g. Paczynski, 1970; Kippenhahn, 1981) and, with the core mass increasing, evolve to higher luminosities along the Hayashi-line (HL) as long as the mass of the hydrogen-rich envelope exceeds a minimum value $M_e \gtrsim 0.05\,M$. The evolutionary track of such a star in the HRD is well approximated by a straight line, i.e. by the relation

$$\log L = \left(\frac{\partial \log L}{\partial \log T_{eff}}\right)_M \log T_{eff} + \left(\frac{\partial \log L}{\partial \log M}\right)_{T_{eff}} \log M + C \tag{9a}$$

or

$$\log L(\mathrm{erg\,s^{-1}}) = a \log T_{eff}(\mathrm{K}) + b \log M(\mathrm{g}) + \mathrm{c}\,. \tag{9b}$$

Together with Stefan-Boltzmann's law, the equilibrium mass radius exponent of models on the HL that obey a core-mass luminosity relation becomes

$$\zeta_{1,e} = 2\frac{\left(\frac{\partial \log L}{\partial \log M}\right)_{HL}}{\left(\frac{\partial \log L}{\partial \log T_{eff}}\right)_{HL}} = \frac{2b}{a}, \tag{10}$$

where the subscript HL indicates that the corresponding derivatives have to be taken along (parallel to) the HL. Results of numerical computations show that $\zeta_{1,e}$ is always negative and typically in the range $-\frac{1}{3} \lesssim \zeta_{1,e} \lesssim -0.2$, depending on the star's chemical composition and the assumed mixing length.

Since, in general, $b \neq 0$, the total radius of an AGB star cannot be independent of the total mass as is often assumed when using a pure core-mass radius relation. Since the position of the HL depends on M, either R or L must depend on M.

Since for the stars in question, the thermal time scale which is of the order of the Kelvin-Helmholtz time

$$\tau_{KH} = \frac{GM^2}{RL}, \tag{11}$$

is much longer than the dynamical time scale

$$\tau_{dyn} = \left(\frac{R^3}{GM}\right)^{1/2} \tag{12}$$

the criterion for dynamical stability against mass transfer (Eq. 1) involves the adiabatic response of the stellar radius to mass loss. Stars on the AGB have a deep outer convection zone and convection deep in the star is very effective and thus very nearly adiabatic. Therefore, one would expect that these stars behave roughly like a polytrope of index $n = \frac{3}{2}$, i.e. that the adiabatic mass radius exponent is

$$\zeta_{1,s} \approx \zeta_{n=3/2} = -\tfrac{1}{3}. \tag{13}$$

However, by approximating AGB stars as polytropes of index $n = \frac{3}{2}$ one ignores the fact that these stars are not fully convective, that the adiabatic temperature gradient in the convection zone is neither constant nor close to 0.4, but lower due to radiation pressure, and that convection in the subphotospheric layers is superadiabatic. Therefore, one would expect that ζ_S of a real AGB star depends on details of its internal structure. While a centrally condensed core in a polytrope with $n = \frac{3}{2}$ leads to an increase of ζ_S over the value $-\frac{1}{3}$ in proportion to the relative core mass M_c/M (Hjellming and Webbink, 1987), the influence of radiation pressure, giving rise to a lower adiabatic temperature gradient and thus to a polytropic index $n > \frac{3}{2}$, and the presence of a subphotospheric, superadiabatic convection zone both tend to make the stellar radius more susceptible to mass loss.

The problem of dynamical stability is further complicated by the fact that the thermal time scale of the outermost layers, of mass M_e, (comprising part or all of the superadiabatic convection zone) is

$$(\tau_{th})_e \approx \frac{GMM_e}{RL} \tag{14}$$

and thus comparable to or shorter than the dynamical time scale if

$$M_e \lesssim \left(\frac{R^3}{GM}\right)^{1/2} \frac{RL}{GM} \approx 0.09\,M_\odot \left(\frac{R}{500\,R_\odot}\right)^{5/2} \left(\frac{L}{10^4\,L_\odot}\right)\left(\frac{M}{M_\odot}\right)^{-3/2}. \tag{15}$$

This means that it is not necessarily correct to take the adiabatic mass radius exponent in Eq. (1), because in very extended stars thermal readjustment of the outer layers is faster than the dynamical readjustment. In the absence of a detailed numerical evaluation of ζ_S, we use, for simplicity, the value $\zeta_S = -\frac{1}{3}$ when estimating the requirements for dynamical stability, but not for the subsequent numerical computations of mass transfer.

Adopting now $\zeta_{1,s} \approx -\frac{1}{3}$ and, for the moment, $k = 0$, we find from Eqs. (1) and (6), or from Fig. 1, that dynamical stability requires $q \lesssim 0.65$, i.e. that the lobe-filling primary has to be much less massive than its companion. This is a well-known result and the ultimate reason why a dynamically stable case C mass transfer is not possible in the classical framework of binary evolution, where at the onset of Roche-lobe overflow the primary is invariably more massive than the secondary.

On the other hand, it is well known that stars on the AGB lose mass via a stellar wind. This mass loss is so strong that it allows stars of an initial mass $M_i \lesssim (8 \pm 2)\,M_\odot$ to become white dwarfs with a mass $M_f < M_{CH} \approx 1.4\,M_\odot$ and $M_f < M_i$ (e.g. Weidemann and Koester, 1983). Therefore, we ask whether there are initial parameters for a binary system such that the primary can lose sufficient mass via a stellar wind on the AGB prior to filling its Roche lobe to allow for a dynamically stable mass transfer.

If mass loss via a stellar wind is taken into account in the stability criterion (Eq. 1), i.e. $k > 0$ in Eq. (6), the critical mass ratio q_{crit} below which mass transfer is dynamically stable becomes smaller still. With $\zeta_{1,S} \approx -\frac{1}{3}$ and $k \approx 1$, i.e. $\dot{M}_{tr} \approx \dot{M}_w$, $q_{crit} \approx 0.5$.

Since at the beginning the primary is the more massive of the two stars, i.e. $q_i > 1$, it has to lose more than half of its initial mass via a stellar wind prior to the onset of mass transfer. Because mass loss of stars of intermediate and low mass, i.e. $1\,M_\odot \lesssim M \lesssim 10\,M_\odot$, becomes effective only on the AGB where the stellar luminosity is high, the primary must, unavoidably, be in an advanced phase of evolution with thermal pulses when mass transfer sets in, in order to meet the above condition. In order to estimate the range of possible initial parameters that allow such a situation to occur, we have to discuss next the evolution on the AGB.

2.2. Steady mass loss of the primary on the AGB

It is clear that the type of binary evolution we envisage here is possible only if the primary can lose a significant fraction (>50%) of its initial mass via steady mass loss on the AGB. A premature loss of its envelope at the base of the AGB via a dynamical ejection or a superwind, as has been discussed for stars with $M_i \lesssim 4\,M_\odot$ (e.g. Mazzitelli and D'Antona, 1986), would prevent the primary from ever filling its Roche lobe.

For the following estimate we shall therefore assume that the primary undergoes steady mass loss via a stellar wind on the AGB before it first fills its critical Roche lobe. The evolution on the AGB can thereby be described approximately in the following way: The luminosity is given by the core-mass luminosity relation. We use the parametrization of Paczyński (1970):

$$L(M_c) = L_*(AM_c - 1)\,, \qquad M_c < A^{-1}\,. \tag{16}$$

The rate $\dot{M}_c$ at which the mass M_c of the degenerate C-O-core grows in time is

$$\dot{M}_c = \frac{L}{X_0 Q_{eff}}\,, \tag{17}$$

where X_0 is the hydrogen mass fraction in the envelope (above the H shell source) and Q_{eff} is the nuclear energy release per unit mass consumed by the hydrogen and helium shell source together.

Following Reimers (1975), mass loss on the AGB is given by

$$\dot{M} = -k_R \frac{LR}{GM} = -1.23\,10^{-5}\,\eta_R \frac{LR}{GM}\,. \tag{18}$$

Equations (16)–(18) together with the relation for the evolutionary track in the HRD (Eq. (9)) and Stefan-Boltzmann's law can be used to evaluate the stellar radius R as a function of mass M. Straightforward integration of these equations yields

$$R(M) = \frac{c^{4/(4-a)}}{(4\pi\sigma)^{a/(4-a)}} \left\{\frac{4(a-b)k_R X_0 Q_{eff}}{(3a-4)GL_*A}\right\}^{(4-a)/(3a-4)} \times \left\{M_i^{2(a-b)/a} - M^{2(a-b)/a}\right\}^{(a-4)/(3a-4)}\,. \tag{19}$$

In the same approximation we can evaluate the core mass $M_{c,f}$ reached at envelope exhaustion from

$$M_{c,f}^{2(a-b)/a} = M_i^{2(a-b)/a} + \frac{4(a-b)c^{2/a}k_R X_0 Q_{eff}}{(3a-4)(4\pi\sigma)^{1/2}GA} \times L_*^{(a-4)/2a}(AM_{c,f} - 1)^{(3a-4)/2a} \tag{20}$$

In order to arrive at specific numbers, we use the following values for the different parameters that enter Eqs. (19) and (20). The Hayashi line (Eq. (9)) is well described by the constants $a = -11.4$, $b = +1.9$ and $c = 13.0$. Using Pacyznski's (1970) core-mass luminosity relation we have $L_* = 3.1\,10^4\,L_\odot$ and $A^{-1} = 0.522\,M_\odot$ in Eq. (16). The hydrogen mass fraction in the envelope is taken as $X_0 \approx 0.7$ and $Q_{eff} \approx 7\,10^{18}\,\mathrm{erg\,g^{-1}}$. With these parameters we obtain the relations $R(M)$ shown in Fig. 2 for two initial masses $M_i = 3\,M_\odot$ and $M_i = 5\,M_\odot$ and for two values of η_R, i.e. $\eta_R = 1$ and $\eta_R = 2$, respectively. The end points of these curves correspond to the point of complete envelope exhaustion and to the corresponding points on the core-mass luminosity relation (Eq. (16)).

2.3. Initial binary parameters compatible with a stable case C evolution

A quick look at Fig. 2 shows that AGB stars that have lost more than half of their initial mass via a steady Reimers-type wind are very extended. Thus the corresponding binary progenitors must be very wide with an orbital separation of the order of $10^3\,R_\odot$ and an orbital period of about a decade.

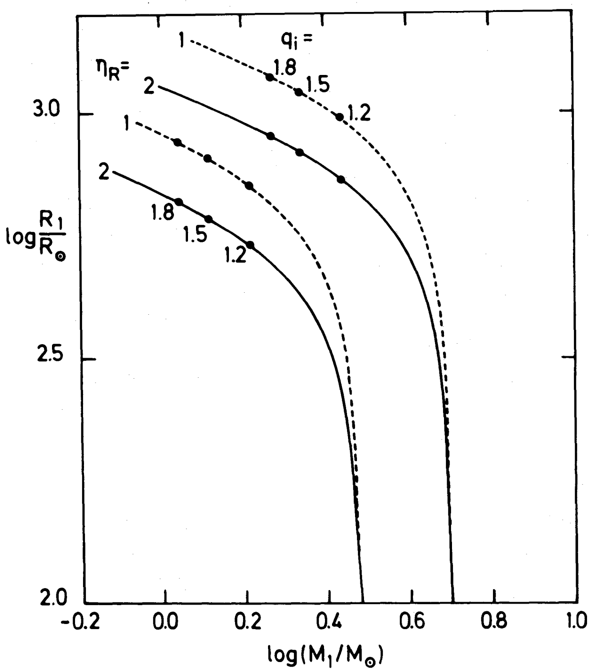

Fig. 2. Radius R_1 of the primary as a function of its mass M_1 during its evolution with mass loss via a Reimers wind (Eq. (18)) on the AGB. The M_1-R_1-relations have been computed from Eq. (19) with $\eta_R = 1$ (dashed lines) and $\eta_R = 2$ (full lines). The curves end at the point where the envelope is exhausted, i.e. where $M_1 = M_c$. At the same time, the end points are given by Eq. (20) and define a core-mass radius relation. The dots along the tracks indicate the highest value of M_1 for which, given the corresponding value of the initial mass ratio q_i, a dynamically stable case C mass transfer is possible

Because of the primary's mass loss via a stellar wind and the associated loss of orbital angular momentum (cf. Eq. (2)) the initial orbital parameters have already changed before the onset of RLOF. Since during the detached phase $\dot{M}_1 = \dot{M}_{1,w}$ and, neglecting wind accretion by the secondary, $\dot{M}_2 = 0$, it follows from Eq. (8) that the change of the primary's Roche radius due to wind mass loss is given by

$$\frac{d\ln R_{1,\mathrm{R}}}{dq} = -\frac{1}{1+q} + \frac{d\ln f_1(q)}{dq}, \tag{21}$$

which, after integration from the initial mass ratio q_i to the current one, yields

$$\ln R_{1,\mathrm{R}}(M_1) = \ln R_{1,\mathrm{R}}(M_i) + \ln\frac{M_{1,i}+M_2}{M_1+M_2} - \ln\frac{f_1\left(\frac{M_{1,i}}{M_2}\right)}{f_1\left(\frac{M_1}{M_2}\right)}. \tag{22}$$

Thus mass loss via a stellar wind from the more massive component widens the system and enlarges the Roche-lobe radius. Only when the primary has lost sufficient mass will further mass loss result in a contraction of the binary.

It is important to note that for all initial mass ratios q_i of interest here, i.e. $1.2 \lesssim q_i \lesssim 1.8$ (see below), the Roche-lobe radius $R_{1,\mathrm{R}}(M)$ increases less steeply than the stellar radius $R(M)$ (as given by Eq. (19)). This means that, given a suitable initial separation, the primary can reach its Roche radius despite the fact that the system widens initially.

Estimating initial binary parameters that are compatible with a stable case C mass transfer proceeds now as follows: First choose the initial mass of the primary $M_{1,i}$ and the initial mass ratio q_i, where two constraints on q_i have to be observed. The first is $q_i > 1$ in order to give the primary the necessary evolutionary lead. If the secondary is required to be still in the phase of central hydrogen burning while the primary is already on the AGB, the dependence of the nuclear time scale on stellar mass yields $q_i \gtrsim 1.2$. The second constraint is that the binary's mass ratio q_f, when RLOF sets in, must be compatible with the dynamical stability criterion, i.e. $q_f < q_{\mathrm{crit}} \approx 0.5 \ldots 0.6$. Thus, the primary's initial mass has to be reduced by stellar winds by a factor $q_i/q_f \approx 2q_i$. The greater q_i, the more mass the primary has to lose on the AGB. If q_i is too high, the envelope will be blown away before $q < q_{\mathrm{crit}}$ is reached and thus no RLOF will occur. A reasonable upper limit for q_i, regarding typical values of the core masses on the AGB, is $q_i \lesssim 1.8$. The choice of q_i then fixes $M_{2,i}$. The initial orbital separation A_i is chosen in such a way that the relation $R_{1,\mathrm{R}}(M_1)$ (Eq. (22)) intersects the relation $R(M_1)$ (Eq. (19)) in the mass range $M_{c,f} < M_1 < q_{\mathrm{crit}} M_2$. This is most easily done graphically as shown in Fig. 3. Typical, numerical values of initial binary parameters are listed in Table 1.

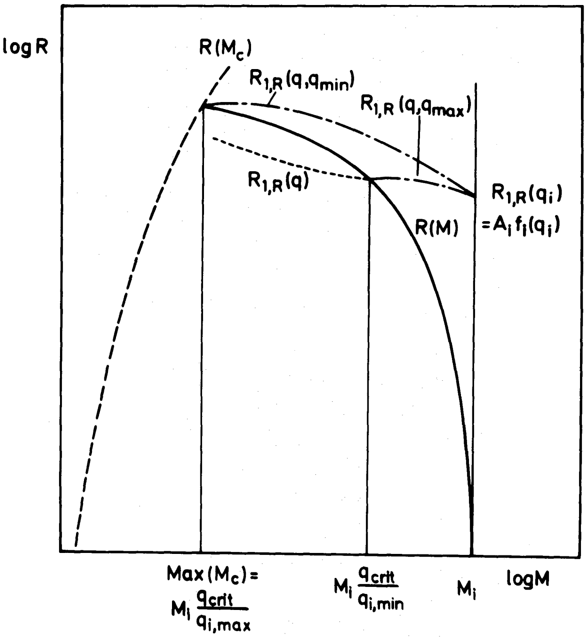

Fig. 3. Scheme for determining the initial binary parameters compatible with a stable case C mass transfer. The full line is the primary's mass radius relation $R_1(M_1)$ according to Eq. (19), also shown in Fig. 2. The dashed line is the core-mass radius relation $R_1(M_c)$. The dashed-dotted lines represent the change of the Roche lobe radius before the onset of mass transfer (due to loss of orbital angular momentum via the stellar wind) according to Eq. (22) for the maximum and minimum initial mass ratio $q_{i,\max}$ and $q_{i,\min}$, respectively. The dotted line sketches the run of the Roche lobe radius after the onset of mass transfer. For further details see the discussion in Sect. 2.3.

3. Numerical computations: I. Standard method

3.1. The stellar evolution program and input physics

We use an updated version of the stellar evolution code described by Hofmeister et al. (1964) and Kippenhahn et al. (1967). In addition to the improvements introduced by Weiss (1986, 1987) we took into account an improved value for the cross section of the helium burning reaction $^{12}C(\alpha,\gamma)^{16}O$ (Kettner et al., 1982) by multiplying the "old" reaction rate by a factor of 3. The opacities used are described in some detail by Weiss (1986, 1987). For the initial chemical composition we took $X = 0.739$, $Y = 0.240$ and $Z = 0.021$, for the ratio of mixing length to pressure scale height $l/H_p = 1.5$. Envelope integrations were carried out from the photosphere down to the fit mass $M_F = 0.97M$. Below M_F convection was assumed to be adiabatic. The algorithm used to follow binary evolution with mass transfer is outlined by Kippenhahn and Weigert (1967) and is hereafter referred to as the standard method.

3.2. The initial model

The proper way of computing the case C evolution discussed in Sect. 2 would have been to choose $M_{1,i}$, $M_{2,i}$ and A_i and to evolve the primary to the point where it first fills its critical Roche lobe, taking into account mass loss via a stellar wind on the AGB e.g. by using Eq. (18). For reasons given below we did not, however, follow this self-consistent approach.

Taking e.g. $M_{1,i} = 3\,M_\odot$ and $q_i = 1.8$, i.e. $M_2 \approx 1.7\,M_\odot$, the primary has to lose $\Delta M_1 \approx 2\,M_\odot$ via a stellar wind before the onset of mass transfer. At the beginning of the AGB phase, i.e. at the first termal pulse, the mass of the degenerate core would roughly be $M_{c,0} \approx 0.65\,M_\odot$. Until the onset of mass transfer the core would grow to $M_c \approx 0.8\,M_\odot$. From the core-mass luminosity relation (Paczyński, 1970) and the core-mass interpulse time

Table 1. Estimated limits of possible orbital parameters of binary systems that undergo a dynamically stable case C mass transfer after the primary has evolved on the AGB and has lost mass via a Reimers-type stellar wind with η_R. A system having a primary of mass $M_{1,i}$ and a mass ratio q_i on the main sequence may begin Roche-lobe overflow (RLOF) at the earliest when M_1, q and A have the values given in columns a), i.e. when $q = q_{crit} \approx 0.65$. The corresponding orbital separation of the system on the main sequence is then A_i. Columns b) give the corresponding parameters for the latest possible onset of RLOF coincident with the exhaustion of the primary's envelope, i.e. when $M_1 = M_c$. For further details see Sect. 2

			a) onset of RLOF when $q = q_{crit}$				b) onset of RLOF when $M_1 = M_c$			
$M_{1,i}/M_\odot$	q_i	η_R	$M_1/M_\odot$	q	$A/10^3R_\odot$	$A_i/10^3R_\odot$	$M_1/M_\odot$	q	$A/10^3R_\odot$	$A_i/10^3R_\odot$
5	1.5	1	2.17	0.65	4.1	2.7	1.21	0.36	3.7	2.0
5	1.2	1	2.71	0.65	4.1	3.1	1.21	0.29	3.5	2.0
3	1.5	1	1.30	0.65	2.8	1.9	0.87	0.44	2.6	1.5
3	1.2	1	1.63	0.65	2.8	2.1	0.87	0.35	2.4	1.5
5	1.5	2	2.17	0.65	3.3	2.2	0.99	0.30	2.9	1.5
5	1.2	2	2.71	0.65	3.3	2.5	0.99	0.24	2.8	1.6
3	1.5	2	1.30	0.65	2.2	1.5	0.75	0.38	2.0	1.1
3	1.2	2	1.63	0.65	2.2	1.7	0.75	0.30	1.9	1.1

relation (Paczyński, 1975) one estimates that in order to reach $M_c \approx 0.8\,M_\odot$ the star has to undergo about 55 thermal pulses. For any other reasonable choice of initial parameters, an analogous estimate shows that before the onset of mass transfer the primary has to go through a large number of thermal pulses. We therefore chose to construct an initial model for our mass transfer computations in a more simple way, not only to save computing time but also to avoid the tedious task of following stellar evolution through a large number of thermal pulses. Instead of following, say, a $3\,M_\odot$ star through 50 thermal pulses we evolved a $5\,M_\odot$ star to the first thermal pulse where the core mass is already $M_c \approx 0.80\,M_\odot$ and thereafter stripped the envelope at an artificially high rate until a suitable total mass was reached. This star then mimics the properties of an initial $3\,M_\odot$ star that went through the required number of thermal pulses.

Of course, we are aware of the fact that this approach is a rather crude one. However, one should keep in mind that the evolution on the AGB is in any case plagued by a number of uncertainties such as whether or not a third dredge-up occurs, what the exact mass loss rate is and, in particular, how it depends on the chemical composition in the photosphere, etc. Thus following the evolution in more detail does not necessarily guarantee that an initial model obtained in this way is closer to reality than one constructed as just described. Since the basic properties of models on the AGB depend, to a very good approximation, only on the mass of the degenerate core, at least as long as the residual envelope mass is $M_e \gtrsim 0.05\,M_\odot$, we are convinced that the basic properties of the subsequent mass transfer phase are not affected by the crude way the starting model has been constructed.

3.3. *The temporal change of the stellar radius of an isolated star during a thermal pulse*

For a better understanding of our numerical results to be described below, it is helpful to discuss briefly how the radius of a single star varies during a thermal pulse. Figure 4a shows the temporal variation of the radius of a $5\,M_\odot$ star over three thermal pulses, while Fig. 4b shows $R(t)$ during the first phases of one thermal pulse but on an extended temporal scale.

After a relatively long phase of stationary double shell source burning the thermal pulse sets in with a thermal runaway in the helium shell source, (A in Fig. 4). As soon as the helium shell source contributes significantly to the total energy generation (which is the case when the thermal instability is already decaying), the stellar radius decreases (A–B in Fig. 4). As a consequence of the rapid expansion of the intershell region, the hydrogen shell source is pushed outwards and extinguishes. At the same time the stellar radius increases (B–C) and reaches a local maximum at point C. Since the helium shell source cannot sustain the total luminosity for long, the star starts contracting (C–D) until the temperature at the layer of the extinguished hydrogen shell source becomes again high enough to ignite hydrogen burning. As soon as the hydrogen shell source again dominates the luminosity production (D), the stellar radius

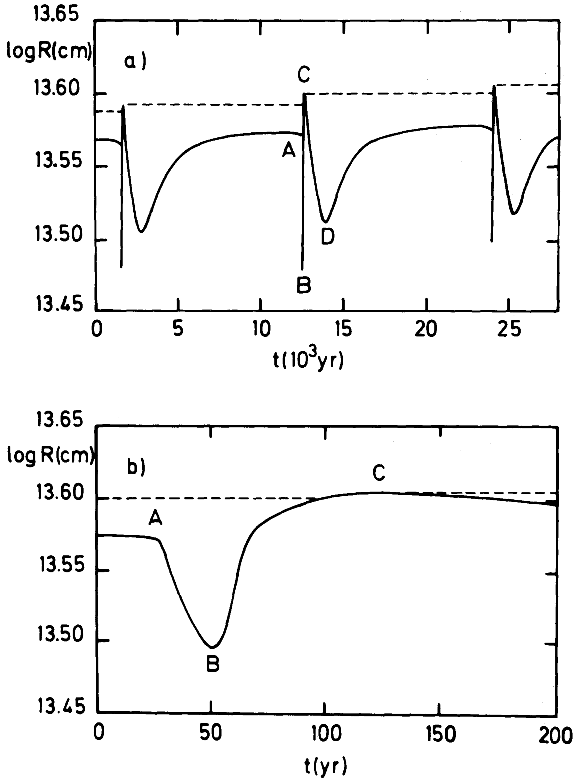

Fig. 4. a The stellar radius as a function of time during the thermal pulses of a $5\,M_\odot$ star with a core mass of $M_c \approx 0.8\,M_\odot$. The interpulse time is 12000 yr. The points labelled A, B, C and D are referred to in Sect. 3.3. **b** Here the phases A-C of the third thermal pulse shown in **a** are shown on an expanded temporal scale

increases until after a prolonged phase of stationary double shell burning the next pulse begins (A).

3.4. *Estimating the main properties of the mass transfer*

It is the rapid increase of the stellar radius between points B and C (cf. Fig. 4) to a maximum that exceeds the radius just before the onset of the pulse, hereafter referred to as the "spike", which deserves our special attention in the context of binary evolution. This is because Roche-lobe overflow will occur only near the tip of the spike. The reason for this is as follows: Let $R_n(\mathrm{C})$ and $R_{n+1}(\mathrm{C})$ be the primary's radius at point C during the thermal pulses n and $n+1$, respectively. Let us further assume that the primary has undergone stable Roche-lobe overflow during the n-th pulse. Therefore, the Roche-lobe radius after the mass transfer episode is $(R_{1,\mathrm{R}}(\mathrm{C}))_n = R_n(\mathrm{C})$. If $R_{n+1}(\mathrm{C}) > R_n(\mathrm{C})$ and if $R_{1,\mathrm{R}}$ increases by less than $R_{n+1}(\mathrm{C}) - R_n(\mathrm{C})$ (e.g. due to wind mass loss) during the interpulse time, the next mass transfer episode will occur during pulse $n+1$ as soon as the primary's radius on the spike (B–C) exceeds $R_n(\mathrm{C})$. During this mass transfer phase the Roche-lobe radius has to increase to the value $R_{n+1}(\mathrm{C})$. After that, mass transfer will stop again at least until the occurrence of pulse $n+2$. The qualitative description given above allows us to obtain a rough estimate of the main characteristics of the mass transfer episodes.

The increase of the Roche-lobe radius on the spike of the pulse implies a total mass transfer of

$$\Delta M_1 = M_1 \zeta_{\mathrm{R},1}^{-1} \Delta \ln R_{1,\mathrm{R}} \tag{23}$$

and a mean mass transfer rate of

$$\langle \dot{M} \rangle \approx M_1 \zeta_{\mathrm{R},1}^{-1} \frac{d \ln R_1}{dt}. \tag{24}$$

Since for typical values of q, $\zeta_{\mathrm{R},1} \approx -\frac{1}{3} \ldots -1$ (cf. Eq. (6) and Fig. 1) and $d \ln R_1/dt \approx 4\,10^{-4}\,\mathrm{yr}^{-1}$ (cf. Fig. 4), we expect

$$\langle \dot{M} \rangle = \langle -\dot{M}_1 \rangle \approx 10^{-3}\,M_\odot\,\mathrm{yr}^{-1} \left(\frac{M_1}{M_\odot}\right)^{-1}, \tag{25}$$

a very high mass transfer rate, though for only a rather short time Δt_{tr}. From Eqs. (23) and (24) we obtain

$$\Delta t_{\mathrm{tr}} \approx \frac{\Delta M_1}{\langle \dot{M} \rangle} = \Delta \ln R_{1,\mathrm{R}} \left(\frac{d \ln R_{1,\mathrm{R}}}{dt}\right)^{-1}, \tag{26}$$

i.e. with $\Delta \log R_{1,\mathrm{R}} = \ln 10[R_{n+1}(\mathrm{c}) - R_n(\mathrm{c})]/R \approx 0.006$ (cf. Fig. 4) $\Delta t_{\mathrm{tr}} \approx 25$ yr. Thus the total amount of mass transferred during one mass transfer episode (one thermal pulse) is $-\Delta M_1 \approx$ few $10^{-2}\,M_\odot$.

3.5. *Results of numerical computations*

We are now in a position to describe the results of the first part of our numerical computations. We followed the evolution of two different systems (hereafter referred to as I and II). The initial parameters of the two systems are listed in Table 2. Each system was evolved under two different assumptions regarding the conservation of mass and angular momentum. On the one hand, we assumed mass transfer to be completely conservative, i.e. we neglect the influence of the stellar wind. In this case, $\dot{J} = 0$ and $\dot{M}_1 + \dot{M}_2 = 0$. On the other hand, we included loss of mass and angular momentum from the system due to the stellar wind. In this case, $\dot{J} = (\dot{J}_{\mathrm{orb}})_{\mathrm{w}}$ (cf. Eq. (2)) and $\dot{M}_1 = \dot{M}_{1,\mathrm{w}} + \dot{M}_{1,\mathrm{tr}}$ with $\dot{M}_2 = -\dot{M}_{1,\mathrm{tr}}$. The orbital separation at the beginning of our computations was chosen such that $R_{1,\mathrm{R}}$ was just slightly larger than R_1 at the peak of the spike during the preceeding (the "first") thermal pulse and mass transfer was initiated with the next one.

3.5.1. Conservative evolution of system I

The evolution after the onset of mass transfer was followed through the subsequent 10 thermal pulses. As an example, Fig. 5 shows the variation in time of the mass transfer rate during the first of the 10 mass transfer episodes. It is seen that mass

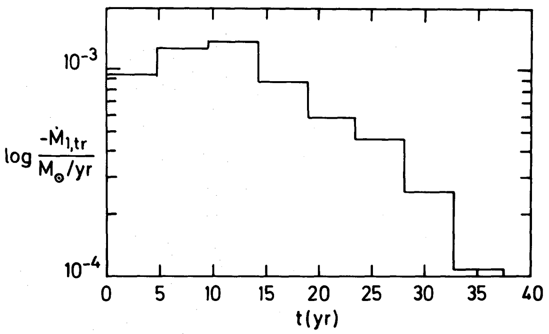

Fig. 5. Mass transfer rate as a function of time during the first thermal pulse in the evolution of system I. The corresponding binary parameters at the onset of mass transfer are listed in Table 2. The step function results from the finite time step (~ 5 yr) used in the computations

Table 2. Orbital parameters of systems I and II at the onset of RLOF and on the main sequence. The main sequence parameters have been estimated with the method outlined in Sect. 2, assuming $\eta_R = 2$ in Eq. (18)

	approximate orbital parameters of the system on the MS						parameters at the onset of ROLF					
System	$M_1/M_\odot$	$M_2/M_\odot$	q	$A/R_\odot$	$R_{1,R}/R_\odot$	P(yr)	$M_1/M_\odot$	$M_2/M_\odot$	q	$A/R_\odot$	$R_{1,R}/R_\odot$	P(yr)
I	3.1	1.71	1.8	900	400	4	1.03	1.71	0.60	1590	530	12.3
II	3.4	2.62	1.3	1200	500	5.5	1.71	2.62	0.65	1660	560	10.4

transfer sets in very abruptly (this is entirely a consequence of the method used for computing $\dot{M}$) with a mass transfer rate of $\sim 10^{-3} M_\odot \, yr^{-1}$. It decreases thereafter by about a factor of 10 over about 40 yr before it drops suddenly to zero. The mass transfer rate as well as the duration of the mass transfer phase agree well with our estimates given in Sect. 3.4. The step function shown in Fig. 5 results from the fact that the mass transfer phase is very short and that the time step used in such computations must not be too small. In the case shown the time step was about 5 yr. However, computations made with half that value did not yield a significantly different result.

The peak and mean mass transfer rates, the duration of the mass transfer episodes and the total mass transferred during each episode of the 10 thermal pulses are shown in Figs. 6a–6c, respectively. It is seen from these Figures that the mass transfer rate (peak and mean value) as well as the duration and the total amount of mass transferred decrease from pulse to pulse. The reason for this behaviour is mainly the decreasing mass ratio, i.e. the decreasing value of $\zeta_{R,1}$ [cf. Fig. 1 and Eqs. (23) and (24)].

3.5.2. Conservative evolution of system II

In this case we followed the evolution through only five mass transfer episodes (thermal pulses). The results we obtained were qualitatively the same as for system I. However, we find that the total amount of mass transferred during one episode is higher than in system I. This is not surprising since M_1, in system II is higher than in system I and according to Eq. (23) $\Delta M_1 \sim M_1$ (while $\Delta \ln R_{1,R}$ is the same because both stars have the same core mass).

3.5.3. Quasi-conservative evolution including mass and angular momentum loss via a stellar wind

The evolution of systems I and II was recomputed, now including the loss of mass and angular momentum via the primary's stellar wind according to Eq. (2) and Eq. (18) with $\eta_R = 1.5$. While the influence of the stellar wind is negligible during the mass transfer episodes, because $-\dot{M}_w \approx$ few $10^{-6} M_\odot \, yr^{-1}$ but $-\dot{M}_{tr} \approx 10^{-3}$–$10^{-4} M_\odot \, yr^{-1}$, its influence on the binary parameters during the interpulse phase is important. Because in our systems I and II $q < q_{crit} < 1$, the stellar wind leads to an increase of the primary's Roche radius. During the interpulse phase, it increases by (cf. Eq. 21)

$$\Delta \ln R_{1,R} = \left(\frac{q}{1+q} - \beta_1 \right) \frac{\tau_{IP}}{\tau_w}, \qquad (27)$$

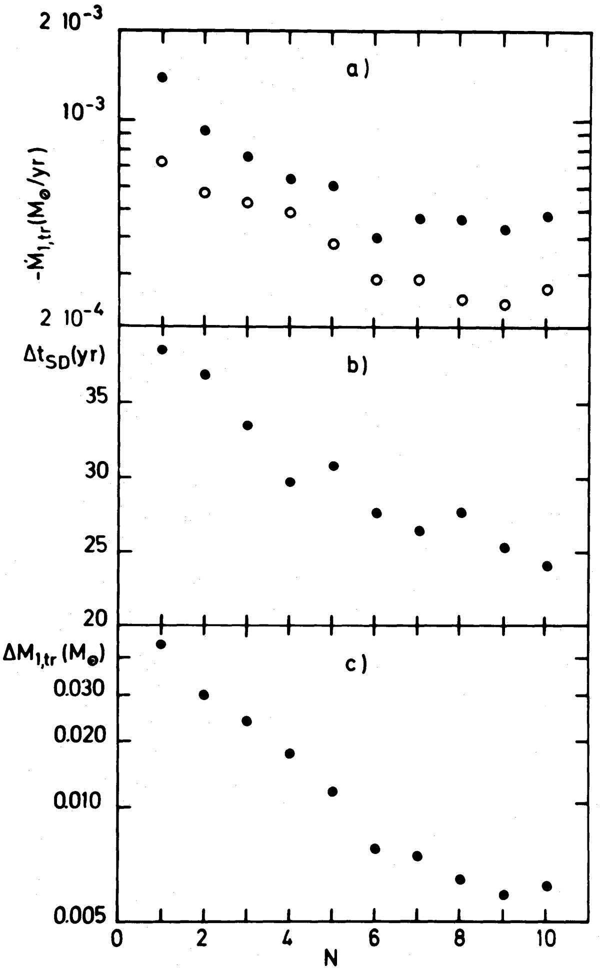

Fig. 6a–c. Characteristic properties of the mass transfer episodes during the 10 thermal pulses in the evolution of system I. **a** Maximum mass transfer rate (full circles) and mean mass transfer rate (open circles). **b** Duration of the individual mass transfer episodes. **c** Total amount of mass transferred during each of the mass transfer episodes

where τ_{IP} is the duration of the interpulse phase (in our case with $M_c \approx 0.8\,M_\odot$ $\tau_{IP} \approx 1.2\,10^4$ yr) and $\tau_w = -M_1/\dot{M}_{1,w} \approx 4\,10^5$ yr is the time scale of mass loss via the stellar wind. With $q \approx 0.6$ and $\beta_1 \approx 0.25$ we obtain $\Delta \log R_{1,R} \approx 0.004$. Because this is of the same order as the difference of the peak radii of two subsequent thermal pulses $\Delta \log R_1 \approx \ln 10(R_{n+1}(C) - R_n(C))/R \approx 0.006$, the mass transfer sets in significantly closer to the peak radius at the next thermal pulse than it would without taking into account the stellar wind. As a consequence, the mass transfer phase is shorter, the mass transfer rate and the amount of mass transferred smaller. Thus the wind mass loss makes it more difficult for the primary to fill its Roche lobe even at the spike of a thermal pulse. In fact, the growth of $R_{1,R}$ during the interpulse phase can exceed the difference $R_{n+1}(C) - R_n(C)$ with the result that no further RLOF will occur. In our computations this happened to system II after the 5th thermal pulse and to system I after the first. The reason why there was only one mass transfer episode in system I is that the primary's envelope was already so close to exhaustion that the wind loss after the first mass transfer phase was sufficient to move the star slightly away from the Hayashi-line to a higher T_{eff} and thus a smaller radius.

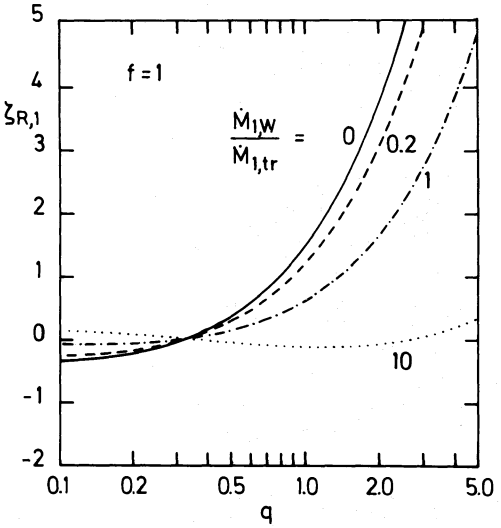

Fig. 7. As Fig. 1 but with $f = 1$ in Eq. (6), i.e. assuming that all the transferred angular momentum remains stored in the accretion disk

3.5.4. Non-conservative evolution

The results discussed so far have shown that this type of case C evolution leads to episodic mass transfer during the spike of a thermal pulse. The mass transfer episodes last for a few decades and are separated by the interpulse time. If we now compare the duration of the mass transfer episodes with the corresponding orbital period, we realize that the mass transfer lasts for only a few orbital periods at most. Therefore, we have to ask ourselves whether the assumption that we have used so far, namely that the transferred angular momentum (cf. Eq. (3)) is instantaneously returned to the orbit, i.e. that we may use Eq. (6) with $f = 0$, is really justified. Thus the question is, on what time scale the transferred angular momentum is returned to the orbit. It is well known that mass flowing from the primary through the inner Lagrangian point L_1 forms an accretion disk around the secondary (see e.g. numerical computations of Hensler, 1982). Once in the disk, matter loses angular momentum via viscous friction and thereby spirals towards the secondary on the viscous time scale of the disk. The angular momentum is thus transferred towards the outer rim of the disk and from there to the orbit via tidal interaction. The time scale on which angular momentum that is stored in the disk is returned to the orbit is thus the same as the one on which matter can spiral to the central object. Unless viscous friction is extremely strong, it takes a particle in the disk many revolutions ($\gtrsim 10^3$) in order to arrive at the centre of the disk. This means that the viscous time scale of the disk is long compared to the orbital period at the outer rim of the disk and thus also long compared to the orbital period of the binary (see the estimate given in Sect. 6, Eq. (39)). Therefore, we must conclude that the transferred angular momentum remains stored in the accretion disk since the viscous time scale is much longer than the duration of a mass transfer episode. This means in turn that f in Eq. (6) is of order unity. Computations of $\zeta_{R,1}$ with $f = 1$ are shown in Fig. 7. It is immediately seen from Fig. 7 that with $\zeta_{1,S} \approx -\frac{1}{3}$, case C mass transfer will be dynamically unstable for all conceivable mass ratios, i.e. $q \gtrsim 0.2$. This is, of course, a rather disappointing result, all the more so since the root of the dynamical instability is not primarily in the stellar components but in the long viscous time scale of the very large accretion disk. Incidentally, this instability is the same as Taam and Mc Dermott (1987) have encountered in the context of cataclysmic binaries. However, the doubts raised against the reality of this instability, at least as far as cataclysmic variables are concerned (Ritter, 1988), are also justified here. Strictly speaking, the instability arises not because of the disk's long viscous time scale, but rather because in our computations mass transfer sets in abruptly. This, in turn, is entirely a consequence of the hidden assumption made for computing the mass transfer rate, namely that the stellar rim of the primary is infinitely sharp. As we shall show in Sect. 5, by relaxing this assumption and taking into account the primary's atmospheric scale height properly, the above-found instability disappears.

4. Computation of the mass transfer rate

4.1. *The standard method and its limitations*

In numerical simulations of binary evolution, the standard method for evaluating the mass transfer rate $\dot{M}_{tr}$ (first described by Kippenhahn and Weigert, 1967) is an iterative procedure subject to the following constraints:

$$\dot{M}_{1,tr} = 0 \quad \text{if} \quad R_1 < R_{1,R} + \varepsilon$$
$$\text{and} \qquad (28)$$
$$\dot{M}_{1,tr} < 0 \quad \text{if} \quad R_1 > R_{1,R} + \varepsilon\,,$$

where $0 < \varepsilon \ll R_{1,R}$ is a small quantity. Inherent in the above prescription, which was also followed in the computations discussed in Sect. 3, is that mass transfer sets in (or turns off) very abruptly (see Fig. 5). The implicit assumption one makes when applying (28) is that the density above the stellar photosphere drops on a scale length small compared to ε. In practice this means that the star has a sharp rim. While the standard method gives reliable results in phases of sustained mass transfer if the stellar surface is sufficiently sharply defined, it is clearly inadequate when mass transfer is not stationary and/or if the atmospheric scale height H is so large that it must be taken into account, i.e. in the framework of (28) if $H \gtrsim \varepsilon$. An appropriate measure for H is the pressure scale height in the photosphere

$$H_{\mathrm{P}} = \left(\frac{dr}{d\ln P}\right)_{r=R} = \frac{R^2 P_0}{GM\rho_0} = \frac{\mathscr{R}TR^2}{\mu GM}, \tag{29}$$

where T is the effective temperature, $\mathscr{R}$ the gas constant, μ the mean molecular weight, and P_0 and ρ_0 are respectively the photospheric pressure and density. It is seen from Eq. (29) that $H_{\mathrm{p}}/R \sim R$, i.e. that the relative thickness of the atmosphere is proportional to the stellar radius. With the typical values of an AGB model, $T_{\mathrm{eff}} \approx 3000\,\mathrm{K}$, $R \approx 500\,R_\odot$, and $\mu = 1.26$, we find $H_{\mathrm{P}}/R \approx 0.05$ or $H_{\mathrm{P}} \approx 25\,R_\odot$. Since the layers above the photosphere are optically thin, they are to a good approximation isothermal. Therefore H_{P} is also the scale height of the density stratification $H_\rho \approx H_{\mathrm{P}}$. This scale length has to be compared with the range of variation of R_1 during one mass transfer episode. In our computations, where $\Delta \log R_1 < 0.006$, we have (with $R_1 = 500\,R_\odot$) $\Delta R_1 \approx 7R_\odot < H_\rho$. Because of this we cannot expect the results of our computations with the standard method to be even qualitatively correct.

Moreover, since the density above the photosphere falls off with such a large scale height, the density at the inner Lagrangian point L_1 and thus the mass flow through the nozzle at the saddle point L_1 become non-negligible long before the photosphere reaches the critical potential.

In short, when dealing with Roche-lobe overflow from stars that have an extended atmosphere it is mandatory to take the finite scale height explicitly into account.

4.2. *Modelling mass transfer as an isothermal flow through L_1*

In order to take the finite scale height of the atmosphere into account, we model mass transfer as an isothermal subsonic flow of gas through L_1 that reaches sound velocity near L_1 (e.g. Plavec et al., 1973; Lubow and Shu, 1975; Meyer and Meyer-Hofmeister, 1983; Pringle, 1985). Here we follow Meyer and Meyer-Hofmeister (1983). Accordingly the mass transfer rate is given by

$$-\dot{M}_{1,\mathrm{tr}} = \rho_{\mathrm{L}_1} v_{\mathrm{s}} Q\,. \tag{30}$$

Here $v_{\mathrm{s}} = (\mathscr{R}T/\mu)^{1/2}$ is the isothermal sound speed, Q the effective cross section and ρ_{L_1} the density of the flow at L_1. ρ_{L_1} is related to the photospheric density ρ_0 via

$$\rho_{\mathrm{L}_1} = \frac{1}{e}\rho_0 e^{-(\Delta R/H_x)^2}\,. \tag{31}$$

In (31) ΔR is the distance of L_1 from the photosphere and H_x is the scale height with which the density above the photosphere falls off in the direction to L_1. Expressions for H_x and Q in terms of the stellar and binary parameters are given in the paper of Meyer and Meyer-Hofmeister (1983) and will not be repeated here.

It is clear that the mass transfer rate computed by means of Eqs. (30) and (31) can only be approximately correct, i.e. within a factor of order unity. This is because a number of simplifications are made in the derivation and application of Eqs. (30) and (31). In particular, when applying (31), it is assumed that the primary is spherical, i.e. that the distance of L_1 from the photosphere is $\Delta R = A_{\mathrm{L}_1} - R_1$, where R_1 is the radius of the (spherical) primary and A_{L_1} the distance of L_1 from the primary's centre of gravity. However, this approximation is not a particularly good one because the tidal deformation of the star, close to filling its critical Roche lobe, is strongest in just that direction. Thus one overestimates ΔR in making this approximation (a better approximation is possible (see e.g. Ritter, 1988) but has not yet been used for the numerical computations to be described below). Furthermore, in deriving (31) and for computing Q only the lowest order expansion of the Roche potential in the vicinity of L_1 is used. Finally it is assumed that the flow is isothermal. Despite these approximations, modelling mass transfer as an isothermal flow through L_1, i.e. applying Eqs. (30) and (31), offers a number of advantages over the standard method. First of all, the finite scale height of the atmosphere is explicitly taken into account, thus allowing for a gradual transition from the detached state (with $\dot{M}_{1,\mathrm{tr}} = 0$) to the semi-detached state with fully developed mass transfer. The abrupt onset or turn-off of mass transfer inherent in the standard method is automatically avoided. The second major advantage is that computing $\dot{M}_{1,\mathrm{tr}}$ is very simple because it is an explicit function of known stellar and binary parameters. Thus the iterative trial and error procedure inherent in the standard method is also avoided.

Of course, computing mass transfer with Eqs. (30) and (31) also has its limitations. In particular it must not be used when $\Delta R < 0$. This means that there is a maximum mass transfer rate $\dot{M}_{\max}$ above which the star must overfill its critical Roche potential and above which $\dot{M}_{\mathrm{tr}}$ must be computed by using a different approximation (e.g. following Paczyński and Sienkiewicz, 1972, or Savonije, 1978). Turning this around, this means that if $-\dot{M}_{1,\mathrm{tr}} < \dot{M}_{\max}$, the photosphere is at a lower potential than the critical value, i.e. in conventional terms the primary underfills it critical Roche lobe and the system would be called detached.

Whether or not the above described method can be applied to the binary systems we are dealing with in this paper depends now on how large $\dot{M}_{\max}$ is. In order to compute $\dot{M}_{\max}$ one has to insert in Eq. (31) $\Delta R = 0$, i.e. $\rho_{\mathrm{L}_1} = (1/e)\rho_0$. This, together with (30) yields (for a derivation see e.g. Ritter, 1988):

$$\dot{M}_{\max} \approx \frac{2\pi}{e}\left(\frac{\mathscr{R}T_{\mathrm{eff}}}{\mu}\right)^{3/2} \frac{R_{1,\mathrm{R}}^3}{GM_1}\rho_0 F(q)\,, \tag{32}$$

where $F(q)$ is a slowly varying function of the mass ratio and is well approximated in the interesting range of q values by

$$F(q) \approx 1.22 - 0.56\log q, \qquad 0.1 \lesssim q \lesssim 2\,. \tag{33}$$

Taking typical values, i.e. $T_{\mathrm{eff}} \approx 3000\,\mathrm{K}$, $\mu \approx 1.3$, $R_{1,\mathrm{R}} \approx 500\,R_\odot$, $M_1 \approx 1\,M_\odot$, $\rho_0 \approx 2\,10^{-10}\,\mathrm{g\,cm^{-3}}$ and $q \approx 0.5$ we find $\dot{M}_{\max} \approx 10^{-4}\,M_\odot\,\mathrm{yr}^{-1}$. Although this is significantly less than the peak transfer rates of $\sim 10^{-3}\,M_\odot\,\mathrm{yr}^{-1}$ encountered in our computations using the standard method (see Sect. 3), our computations including Eqs. (30) and (31) (to be described in the next section) show that $10^{-4}\,M_\odot\,\mathrm{yr}^{-1}$ is more than adequate. In fact, in none of our computations was the peak transfer rate higher than $2\,10^{-5}\,M_\odot\,\mathrm{yr}^{-1}$.

5. Numerical results: II. The alternative method

5.1. *The starting model*

Because of the rather large atmospheric scale height of the primary $H_{\mathrm{P}} \approx 0.05\,R$, mass transfer through L_1 is already non-negligible when the photosphere lies still deep, i.e. a few times H_{P}, below the critical potential surface. A realistic model of the evolution in question would therefore require that the computations be started sufficiently early in the system's evolution. However, the approximations used to compute $\dot{M}_{1,\mathrm{tr}}$ [Eqs. (30) and (31) in Sect. 4] are not adequate if $\Delta R/H_x \gtrsim 1 \ldots 2$. Therefore, in this paper we restrict ourselves to computations where mass transfer is switched on rather late in the evolution when $\Delta R/H_x \lesssim 1 \ldots 2$. Nevertheless, these computations are suitable for illustrating the main point we want to make here, namely that the results obtained in this way are qualitatively different from those using the standard method. More realistic calculations, where the turn-on of mass transfer is treated more adequately require a slightly more general formalism than that described in Sect. 4, though along exactly the same lines. Such computations will be presented in a subsequent paper.

As has been mentioned above, we wish to show with these computations how mass transfer proceeds when modelled as an isothermal flow through L_1. To illustrate the case in point we followed the evolution of two different systems (hereafter referred to as system III and system IV). The starting values of the two differ only in the initial orbital separation A_i (see Table 3). For system III, A_i was chosen such that at the beginning of mass transfer $\dot{M}_{1,\mathrm{tr}} \approx 0.1\,\dot{M}_{1,\mathrm{w}}$, i.e. that mass loss via a stellar wind was dominant. For system IV, on the other hand, A_i was taken slightly smaller such that initially $\dot{M}_{1,\mathrm{tr}} \approx \dot{M}_{1,\mathrm{w}}$.

5.2. *Quasi-conservative evolution*

Here we discuss the results of computations where we assumed that the angular momentum transferred to the accretion disk is returned instantaneously to the orbit, i.e. where $f = 0$ in Eq. (6). In the light of our discussion in Sect. 3.5 this might seem to be a questionable assumption. However, we shall show below (Sect. 5.3) that in this case the assumption is well justified.

The starting parameters of system III are listed in Table 3. The evolution of this system was then followed through six thermal pulses of the primary. The variations with time of $\dot{M}_{1,\mathrm{tr}}$ and $\dot{M}_{1,\mathrm{w}}$ are shown in Fig. 8a, those of R_1 and $R_{1,\mathrm{R}}$ in Fig. 8b. It is seen that at the beginning $-\dot{M}_{1,\mathrm{tr}} \approx 4\,10^{-7}\,M_\odot\,\mathrm{yr}^{-1}$ and $-\dot{M}_{1,\mathrm{w}} \approx 3\,10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$. Both rates increase with time, though the mass transfer rate accelerates faster. At the end of the computations $-\dot{M}_{1,\mathrm{tr}} = 2.5\,10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$ and $-\dot{M}_{1,\mathrm{w}} \approx 4.5\,10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$. The corresponding binary parameters are listed in Table 4. The evolution, as far as has been computed, is thus dominated by mass loss via the stellar wind. In fact, the total mass transferred during the 66000 yr is only $\Delta M_{\mathrm{tr}} = 0.07\,M_\odot$, while the wind loss over the same time is $\Delta M_{\mathrm{w}} = 0.22\,M_\odot$.

The initial parameters of system IV are also listed in Table 3. As has already been mentioned, these parameters differ from those of system III only in the smaller orbital separation of system IV. Accordingly, $\dot{M}_{1,\mathrm{w}}$ at the beginning of the evolution was the same as in system III, i.e. $-\dot{M}_{1,\mathrm{w}} \approx 3\,10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$. Because of the smaller value of A_i, however, $-\dot{M}_{1,\mathrm{tr}}$ is now higher, namely $-\dot{M}_{1,\mathrm{tr}} \approx 2\,10^{-6}\,M_\odot\,\mathrm{yr}^{-1}$. The subsequent evolution was followed through 8 thermal pulses. $\dot{M}_{1,\mathrm{tr}}$ and $\dot{M}_{1,\mathrm{w}}$ as a function of time are shown in Fig. 9a, those of R_1 and $R_{1,\mathrm{R}}$ in Fig. 9b, respectively. It is seen from Fig. 9a that the temporal mean values of both mass loss rates (taken over one thermal pulse) increase throughout the evolution and this despite the fact that after the 6$^{\mathrm{th}}$ pulse R_1 starts decreasing while $R_{1,\mathrm{R}}$ and thus ΔR begin to grow significantly. Why then does the transfer rate nevertheless grow? The main reason for this is that not only ΔR increases but H_x does so even more so that $\Delta R/H_x$ in Eq. (31) becomes smaller.

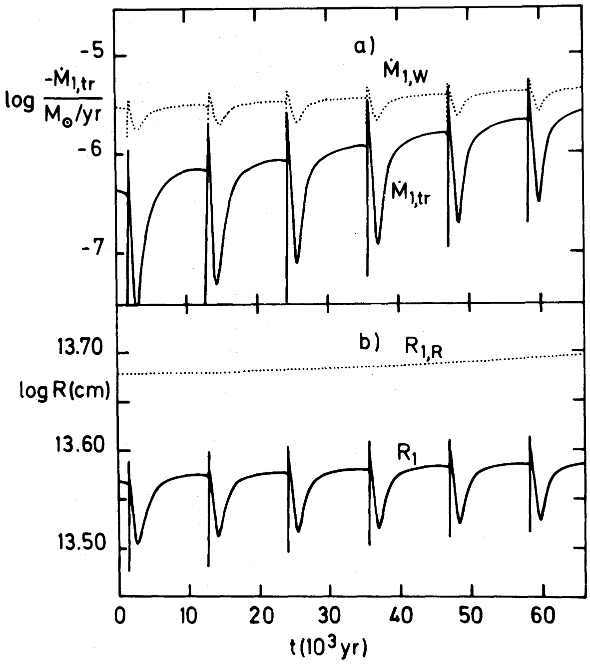

Fig. 8a and b. Evolution of system III. **a** Mass transfer rate (full line) and wind loss rate (dotted line) as a function of time. The corresponding binary parameters at the beginning and at the end of the computations are listed, respectively, in Tables 3 and 4. **b** Primary's radius R_1 (full line) and the corresponding Roche lobe radius $R_{1,\mathrm{R}}$ (dotted line) as a function of time

Table 3. Orbital parameters of systems III and IV at the beginning of the computations of mass transfer according to the method outlined in Sect. 4

System	$M_1/M_\odot$	$M_2/M_\odot$	q	P(yr)	$A/10^2R_\odot$	$R_1/10^2R_\odot$	$R_{1,R}/10^2R_\odot$	$\Delta R/10^2R_\odot$	$H_x/10^2R_\odot$	$R_D/10^2R_\odot$
III	1.71	2.62	0.65	13.9	20.3	5.3	6.9	3.9	1.5	5.9
IV	1.71	2.62	0.65	10.9	17.3	5.3	5.9	2.6	1.2	5.1

Table 4. Orbital parameters of systems III and IV at the end of the computations of mass transfer according to the method outlined in Sect. 4

System	$M_1/M_\odot$	$M_2/M_\odot$	q	P(yr)	$A/10^2 R_\odot$	$R_1/10^2 R_\odot$	$R_{1,R}/10^2 R_\odot$	$\Delta R/10^2 R_\odot$	$H_x/10^2 R_\odot$	$R_D/10^2 R_\odot$
III	1.41	2.69	0.53	16.4	22.2	5.5	7.2	4.1	1.8	6.8
IV	0.90	3.07	0.29	24.0	28.3	5.1	8.0	5.7	2.7	9.7

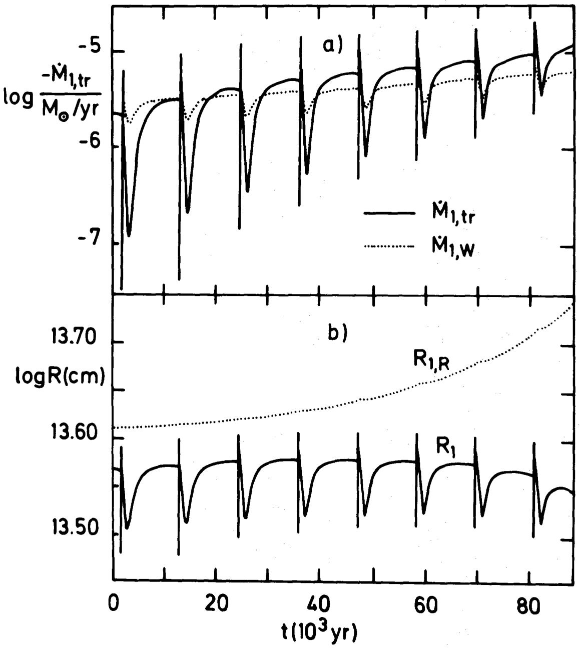

Fig. 9a and b. Evolution of system IV. **a** Mass transfer rate (full line) and wind loss rate (dotted line) as a function of time. The corresponding binary parameters at the beginning and at the end of the computations are listed, respectively, in Tables 3 and 4. **b** Primary's radius R_1 (full line) and the corresponding Roche lobe radius $R_{1,R}$ (dotted line) as a function of time

The significant increase in H_x, in turn, is a consequence of the system's changing mass ratio, as can be seen from the final binary parameters listed in Table 4. The reason why R_1 starts decreasing after the 6th thermal pulse is that the mass of the hydrogen-rich envelope is already so small ($0.1\,M_\odot$ above a $0.8\,M_\odot$ C-O-core at the end of the computation) that the primary begins to move away from the Hayashi-line towards higher effective temperatures.

The most important aspect of these simulations is, however, that the results are qualitatively different from those obtained by using the standard method for computing $\dot M_{1,\rm tr}$. When the finite scale height of the atmosphere is adequately taken into account, we find the binary evolving with sustained mass transfer while the standard method yields episodic mass transfer.

Although mass transfer is sustained throughout the evolution (at least as far as the computations go), it is by no means monotonic. Rather $\dot M_{1,\rm tr}$ is strongly modulated due to the variations of R_1 during a thermal pulse (see Figs. 8 and 9). The mass transfer rate changes particularly strongly during the spike of a thermal pulse and the associated time scale is that of the spike, i.e. a few decades at most (see Fig. 4b). Therefore, we encounter once more the problem discussed in Sect. 3.5, namely whether the assumption that the transferred angular momentum is given back to the orbit sufficiently rapidly is still justified, and, if this is not the case, whether the mass transfer will be dynamically stable. With this end in view we have performed a simulation in which we assume that all the transferred angular momentum remains stored in the disk (at least for a finite amount of time).

5.3. *Non-conservative evolution*

In order to study the consequences of keeping the transferred angular momentum stored in the accretion disk, we have performed one simulation where, starting from a model of sequence IV before the 5th thermal pulse, mass transfer was computed by taking $f = 1$ in Eq. (6). The results of this simulation which covers about 10^4 yr are shown, respectively, in Figs. 10a and 10b. For comparison the results obtained for the same sequence with $f = 0$ are also shown in Fig. 10. It is seen that over the $\sim 10^4$ yr the evolution with $f = 1$ is qualitatively the same as with $f = 0$. Because of the additional angular momentum sink in the system when $f = 1$, the system contracts slowly, ΔR decreases and, therefore, $\dot M_{1,\rm tr}$ accelerates slightly faster. It is clear that when the evolution with $f = 1$ is continued over a time much longer than the $\sim 10^4$ yr considered here, the system will eventually run into the same difficulties as we encountered with the standard method: mass transfer is dynamically unstable and the mass transfer rate will run away.

In reality, however, the amount of orbital angular momentum lost due to mass transfer at a given time is (see Eq. (3))

$$\dot J_{\rm orb}(t)_{\rm tr} = f X^2_{L_1,M_2} A^2 \omega [\dot M_{1,\rm tr}(t) - \dot M_{1,\rm tr}(t - \tau_{\rm visc})] , \tag{34}$$

where $\tau_{\rm visc}$ is the viscous time scale of the accretion disk, i.e. the time scale on which the angular momentum stored in the disk is returned to the orbit. Therefore, only if $\dot M_{1,\rm tr}$ is modulated on a time scale that is short compared to $\tau_{\rm visc}$ is the disk an effective sink (or source) of angular momentum. This is e.g. the case when mass transfer sets in abruptly as in our simulations using the standard method.

With regard to our simulation where we kept $f = 1$ for $\sim 10^4$ yr, we must now ask whether the diffusion (viscous) time scale of the disk is longer or shorter than $\sim 10^4$ yr. If it is not much longer, our worst case simulation is sufficient to show that the mass transfer does not run away because of temporal angular

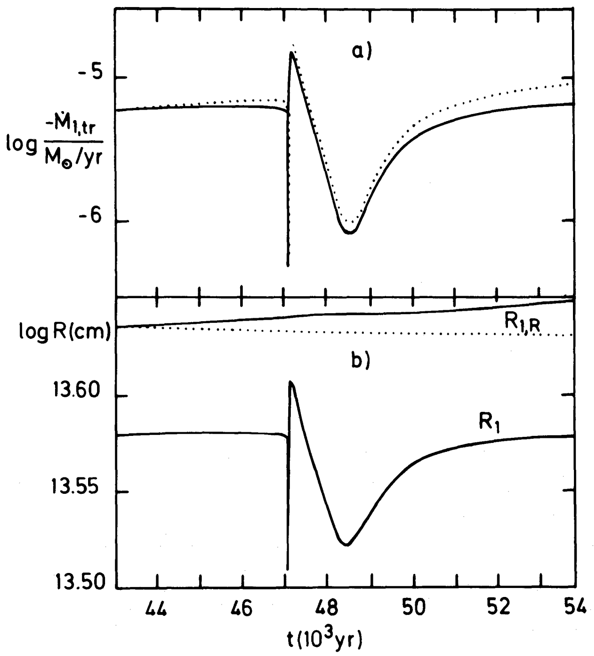

Fig. 10a and b. Evolution of system IV beginning with the 5th thermal pulse. Comparison of the evolution with, respectively, $f = 0$ and $f = 1$ in Eq. (6). **a** Mass transfer rate as a function of time with $f = 0$ (full line) and $f = 1$ (dotted line). **b** Stellar radius R_1 (lower full line) and the corresponding Roche-lobe radius $R_{1,R}$ with $f = 0$ (upper full line) and $f = 1$ (dotted line) as a function of time

momentum storage in the disk. In fact, as our discussion in the following section of the properties of accretion disks will show, τ_{visc} is very probably shorter than 10^4 yr in the disks of interest. Because the total amount of angular momentum stored in the disk

$$J_{disk} \approx \int_{\tau_{visc}} f X^2_{L_1,M_2} A^2 \omega \dot{M}_{tr}\, dt \tag{35}$$

is always small compared to the orbital angular momentum as long as the mass of the disk $M_{disk} \approx \dot{M}_{tr}\tau_{visc} \ll M_1 + M_2$, the disk, as a sink of angular momentum, cannot secularly destabilize mass transfer in a system that is dynamically stable against conservative mass transfer ($f = 0$) and where the lobe-filling component actually underfills its critical Roche-lobe (see also the discussion of the same problem in the context of cataclysmic binaries by Ritter, 1988).

6. The accretion disk and the accreting secondary

6.1. The accretion disk

Here we wish to discuss briefly the main properties of the accretion disk that forms around the secondary as a consequence of mass transfer. These properties are of interest here for a number of reasons. On the one hand, with regard to our discussion in Sect. 5.3, we would like to know the viscous time scale of the disk. On the other, the properties of the accretion disk are to a certain extent important for the observational appearance of this type of binary system.

First of all, it is important to realize that the disk in such a system is unlike the more familiar ones prevailing in cataclysmic variables. The reason for this is the disk's enormous size. Its outer, tidally truncated radius is

$$R_D \approx 0.7 R_{2,R} \approx 0.7 q^{-0.45} R_{1,R}\,. \tag{36}$$

Inserting typical values for $R_{1,R}$ and q we arrive at values $R_D \approx 500\, R_\odot$. Given the typical accretion rates of 10^{-5}–$10^{-6}\, M_\odot\,yr^{-1}$ (see Sect. 5 and Figs. 8–10), the disk is expected to be very cool, at least in its outer parts. Assuming for the moment a stationary optically thick disk whose surface temperature is determined by viscous heating only, the effective temperature at the outer rim is then given by (e.g. Smak, 1984)

$$T_{eff}(R_D) \approx \left(\frac{3}{8\pi\sigma}\frac{GM_2\dot{M}}{R_D^3}\right)^{1/4}$$
$$\approx 140\,K \left(\frac{M_2}{M_\odot}\right)^{1/4}\left(\frac{\dot{M}}{10^{-6}\,M_\odot\,yr^{-1}}\right)^{1/4}\left(\frac{R_D}{500\,R_\odot}\right)^{-3/4}. \tag{37}$$

Even irradiation of the disk by the luminous (but rather cool) primary with $T_{eff} \approx 3000$ K cannot raise the disk's effective temperature above $\sim 10^3$ K at the outer rim. Therefore, the outer parts of the disk are very cool and we must expect dust formation in regions where $T_D \lesssim 10^3$ K.

Another property of importance is the vertical goemetrical thickness of the disk. Hydrostatic equilibrium in the vertical direction demands a vertical pressure scale height H_D, where (e.g. Meyer, 1986)

$$\frac{H_D}{R_D} = \left(\frac{2\mathcal{R}T_D R_D}{\mu G M_2}\right)^{1/2}$$
$$\approx 0.2\mu^{-1/2}\left(\frac{T_D}{10^3\,K}\right)^{1/2}\left(\frac{M_2}{M_\odot}\right)^{-1/2}\left(\frac{R_D}{500\,R_\odot}\right)^{1/2}. \tag{38}$$

Because of the relatively large geometrical thickness at its outer radius the disk cannot be irradiated effectively by the primary. Therefore the surface temperature of the disk's outer parts is likely to be lower than $\sim 10^3$ K.

The viscous time scale of the disk is given by (e.g. Meyer, 1986)

$$\tau_{visc} \approx \left(\frac{R_D}{H_D}\right)^2 \frac{1}{\alpha\Omega_K(R_D)} = \frac{1}{\alpha}\frac{\mu}{2\mathcal{R}T_D}(GM_2R_D)^{1/2}$$
$$\approx 13\,yr\,\frac{\mu}{\alpha}\left(\frac{M_2}{M_\odot}\right)^{1/2}\left(\frac{R_D}{500\,R_\odot}\right)^{1/2}\left(\frac{T_D}{1000\,K}\right)^{-1}. \tag{39}$$

Here $\Omega_K(R_D)$ is the Keplerian angular frequency at the disk's outer radius and α the dimensionless viscosity parameter introduced by Shakura and Sunyaev (1973), i.e. the ratio of the viscous stresses to the gas pressure. Because of the upper limit of $\alpha \lesssim 1$, Eq. (39) yields a lower limit of τ_{visc}. In our case it is of the order of one to a few decades. This means that τ_{visc} is at least of the order of the orbital period and comparable to (or longer than) the duration of the mass transfer episodes discussed in Sect. 3. A priori, the exact values of α (and its possible dependence on H/R) are not known. For the small and relatively hot disks as found in dwarf novae and low-mass X-ray binaries ($R_D \approx R_\odot$, $T_D \gtrsim 10^4$ K), α can be calibrated empirically from the decay time of dwarf nova outbursts. Typical values of α necessary to account for the observed decay time are in the range $0.1 \lesssim \alpha \lesssim 1$. On the other hand, if one applies the disk instability model to the outbursts of symbiotic stars where the accretion disk is much larger, i.e. $R_D \approx 50\,R_\odot$, one arrives at $\alpha \approx 0.05$ (Duschl, 1986). The disks we are dealing with here are larger still (by about another factor of 10) and certainly much cooler in their outer parts. This leads to the suspicion that α could be even smaller, perhaps as small as 10^{-3}. Taking for the moment the pessimistic view that $\alpha = 10^{-3}$, our above estimate of the viscous time scale (Eq. (39)) yields $\tau_{\rm visc} \approx 10^4$ yr. This is the value we had in mind when performing the non-conservative simulation described in Sect. 5.3.

Given an estimate of α, i.e. of the viscosity, we can estimate the total mass stored in a stationary disk

$$M_D \approx \dot{M}\tau_{\rm visc} \approx 2\,10^{-5}\,M_\odot \frac{\mu}{\alpha}\left(\frac{M_2}{M_\odot}\right)^{1/2}\left(\frac{R_D}{500\,R_\odot}\right)^{1/2}\left(\frac{T_D}{10^3\,\rm K}\right)^{-1} \times \left(\frac{\dot{M}}{10^{-6}\,M_\odot\,\rm yr^{-1}}\right). \tag{40}$$

This shows that for typical values of R_D, T_D and $\dot{M}$, and unless $\alpha < 10^{-3}$, the mass of the disk remains small compared to M_2 and thus to $M_1 + M_2$. As a consequence the angular momentum stored in the disk is also small compared to the orbital angular momentum of the binary.

Furthermore, we would like to know whether such a cool disk is optically thick in the vertical direction (as we have assumed so far). For an estimate, we assume again that the disk is stationary. The optical depth of the disk perpendicular to the orbital plane is

$$\tau = \int_{-\infty}^{\infty} \kappa\rho\,{\rm d}z = \bar{\kappa}\int_{-\infty}^{\infty}\rho\,{\rm d}z = \bar{\kappa}\Sigma\,, \tag{41}$$

where κ is the opacity per unit mass, ρ the density and Σ the surface density of the disk. For a stationary disk we have

$$\Sigma = \frac{\dot{M}}{2\pi R_D v_R} \tag{42}$$

where

$$v_R \approx \frac{R_D}{\tau_{\rm visc}} = R_D\left(\frac{H_D}{R_D}\right)^2 \alpha\Omega_K(R_D) = \alpha\frac{2\mathcal{R}T_D}{\mu}\left(\frac{R_D}{GM_2}\right) \tag{43}$$

is the radial drift velocity due to viscous angular momentum transport. Equations (41)–(43) together yield

$$\tau = \frac{1}{\alpha}\frac{\bar{\kappa}\mu\dot{M}}{4\pi\mathcal{R}T_D}\left(\frac{GM_2}{R_D^3}\right)^{1/2} \approx 3.4\frac{\mu}{\alpha}\left(\frac{M_2}{M_\odot}\right)^{1/2}\left(\frac{R_D}{500\,R_\odot}\right)^{-3/2} \times\left(\frac{T_D}{10^3\,\rm K}\right)^{-1}\left(\frac{\dot{M}}{10^{-6}\,M_\odot\,\rm yr^{-1}}\right)\left(\frac{\bar{\kappa}}{\rm cm^2\,g^{-1}}\right). \tag{44}$$

Since the temperature at the outer rim of the disk is so low, the dominant source of opacity is dust. In the relevant temperature and density regime we have typically (e.g. Alexander, 1975) $\bar{\kappa} \approx 1\,{\rm cm^2\,g^{-1}}$. Thus it follows from (44) that at least the outer parts of the disk are also optically thick.

Finally we are also interested in some of the disk's properties near the central object. In the framework of this paper, the latter is a main sequence star with a mass of a few $M_\odot$ (see Table 1, where $M_2 = M_1/q$). Thus its radius is $R_2 \approx 1$–$2\,R_\odot$ (however, see also below, Sect. 6.2). If the disk accretes stationarily and extends all the way down to the secondary's surface, the maximum effective temperature in the disk $T_{\rm eff,max}$ is reached at a distance $R_{\rm max} = 49/36 R_{\rm i}$ where $R_{\rm i}$ is the disk's inner radius and (e.g. Frank et al., 1985)

$$T_{\rm eff,max} = \left\{\frac{3GM_2\dot{M}}{8\pi\sigma R_{\rm max}^3}\left(1-\left(\frac{R_2}{R_{\rm max}}\right)^{1/2}\right)\right\}^{1/4} = \left(\frac{6}{7}\right)^{7/4}\left(\frac{GM_2\dot{M}}{16\pi\sigma R_2^3}\right)^{1/4} = 7.37\,10^3\,{\rm K}\left(\frac{M_2}{M_\odot}\right)^{1/4}\left(\frac{R_2}{R_\odot}\right)^{-3/4}\left(\frac{\dot{M}}{10^{-6}\,M_\odot\,\rm yr^{-1}}\right)^{1/4}. \tag{45a}$$

If, on the other hand, the central object rotates critically, $T_{\rm eff,max}$ is reached at $R_{\rm i}$. With $R_{\rm i} \approx R_2$ we have

$$T_{\rm eff,max} = \left(\frac{GM_2\dot{M}}{8\pi\sigma R_2^3}\right)^{1/4} = 1.15\,10^4\,{\rm K}\left(\frac{M_2}{M_\odot}\right)^{1/4}\left(\frac{R_2}{R_\odot}\right)^{-3/4}\left(\frac{\dot{M}}{10^{-6}\,M_\odot\,\rm yr^{-1}}\right)^{1/4}. \tag{45b}$$

Much higher temperatures are, however, reached in the boundary layer that is formed between the disk and the non-rotating star. Assuming that the energy dissipated in the boundary layer is radiated as blackbody radiation over an area of $2\pi R_2 \cdot 2H_{\rm BL}$, where

$$H_{\rm BL} = \frac{T_{\rm BL}R_2^2}{\mu GM_2} \tag{46}$$

is the pressure scale height of the boundary layer in the radial direction (see Frank et al., 1985) we arrive at a boundary layer temperature of

$$T_{\rm BL} = \left(\frac{G^2\mu M_2^2\dot{M}}{8\pi\sigma R_2^4}\right)^{1/5} = 4.7\,10^4\,{\rm K}\mu\left(\frac{M_2}{M_\odot}\right)^{2/5}\left(\frac{R^2}{R_\odot}\right)^{-4/5} \times\left(\frac{\dot{M}}{10^{-6}\,M_\odot\,\rm yr^{-1}}\right)^{1/5}. \tag{47}$$

Thus from Eqs. (45) and (47) we may conclude that the disk is relatively cool, i.e. $T \lesssim 10^4$ K almost everywhere, except for the innermost region near the central object, where it might or might not be hot enough (depending among other things on the secondary's spin frequency) for disk instabilities to occur (see e.g. Smak, 1984; Duschl, 1986). The bolometric luminosity of the accretion disk is

$$L_D = \frac{GM_2\dot{M}}{2R_2} = 15.7\,L_\odot\left(\frac{M_2}{M_\odot}\right)\left(\frac{R_2}{R_\odot}\right)^{-1}\left(\frac{\dot{M}}{10^{-6}\,M_\odot\,\rm yr^{-1}}\right). \tag{48}$$

If the secondary does not rotate, the rotational energy of the accreted matter is dissipated in the boundary layer giving rise to the luminosity $L_{\rm BL} = L_D$. Whether or not the boundary layer does contribute, the total luminosity (at most $2L_D$) is always much smaller (by a factor of $10^2 \ldots 10^3$) than the luminosity of the primary star. This is a consequence of the relatively large inner disk radius $R_{\rm i} \approx R_2 \approx R_\odot$ and of the moderate mass transfer rates in the systems in question. Therefore, the disk will manifest itself primarily by its obscuration and not by its radiation, i.e. by eclipsing the primary if the orbital inclination is higher than about 70°. Because of the disk's enormous size when compared to the primary, i.e. $R_D = 1$–$2R_1$ (see Tables 3 and 4), and because H_D/R_D is not small, the eclipses can be very long, i.e. $\Delta\varphi_{\rm Ecl} = 2$ arcsin $(R_D + R_1)/A = 60°$–$70°$ in orbital phase, and of rather large amplitude, i.e. $\Delta m = 0.3$–0.5 mag.

6.2. The accreting secondary

Here we wish to discuss briefly to what extent the properties of the secondary star are altered as a consequence of the enforced accretion from the primary at rates of the order 10^{-5}–$10^{-6}\,M_{\odot}\,\mathrm{yr}^{-1}$. The consequences of mass accretion on main sequence stars have been investigated in some detail by Neo et al. (1977) and Kippenhahn and Meyer-Hofmeister (1977). The basic result of these two studies was that a main sequence star with a radiative envelope, i.e. with $M \gtrsim M_{\odot}$, will expand upon accretion if the accretion time scale $\tau_M = M/\dot{M}$ is shorter than or comparable to the Kelvin-Helmholtz time (Eq. (11)) of the accretor. Since the latter is of the order of 10^7 yr for the stars in question whereas $\tau_M \approx 10^5$–10^6 yr, the secondary's reaction could be significant. However, based on the numerical results of the papers quoted above, we find in fact that the secondary could double its radius at most (cf. the evolution of a $2\,M_{\odot}$ star accreting at $2\,10^{-5}\,M_{\odot}\,\mathrm{yr}^{-1}$ in Kippenhahn and Meyer-Hofmeister, 1977). Though not negligible, the secondary's reaction will still not change the qualitative picture outlined above. In particular, it is clear that the system will remain semi-detached since the accretion rates are far too small for the secondary to expand to the point where a contact or even an over-contact system (that could give rise to a spiralling-in evolution) is formed.

A potentially more serious problem for the secondary's internal structure derives from the fact that it accretes not only mass but also angular momentum and is thus spun up. Packet (1981) has stressed that surprisingly little mass (less than 10% of the accretor's initial mass) need be accreted before the star is spun up to break-up velocity. As our computations in Sect. 5 show, the amount of mass transferred can easily exceed $0.1\,M_{\odot}$ and thus give rise to an angular momentum problem for the secondary.

7. Discussion

Given the results we have obtained so far, we may now ask how they depend on the initial mass of the primary on the main sequence. From our estimates in Sect. 2 (see also Figs. 2 and 3) it is clear that with increasing $M_{1,i}$ the orbital separation and thus the period must increase as well, while the initial mass ratio remains constrained as before. Correspondingly, at the onset of RLOF, the mass of the primary, its core mass, the orbital separation and the period will be larger. Because of the higher core mass, the primary's luminosity is higher and its radius larger. As a result, both the mass transfer rate and the wind loss rate will be higher too. However, since $\dot{M}_{1,\mathrm{tr}} \sim (\partial \ln R_1/\partial t) \sim (d \ln R_1/d \ln M_c) L_1/M_c$ and $\dot{M}_{1,\mathrm{w}} \sim L_1 R_1/M_1$, the ratio of the wind loss rate to the mass transfer rate $\dot{M}_{1,\mathrm{w}}/\dot{M}_{1,\mathrm{tr}} \sim M_c R_1/M_1$ is increasing with M_c and thus with $M_{1,i}$. Thus the higher $M_{1,i}$, i.e. M_c, the more important the wind loss becomes as compared to the mass transfer. This is in agreement with numerical results obtained by de Kool et al. (1986).

Finally we may ask what the ultimate fate of these systems will be. The immediate outcome is clear: After the primary has lost its hydrogen-rich envelope it will become a white dwarf. The binary will thus end as a long-period detached system consisting of a main sequence star of a few $M_{\odot}$ and of a white dwarf. Although these are the hallmarks of systems like Sirius AB and Procyon AB there are reasons to believe that these two systems have formed in a different way (see e.g. D'Antona, 1982 for a discussion). The question whether the subsequent evolution of the secondary will lead to a second RLOF later on cannot be answered without a more detailed analysis. Since the orbital separation is very large it is not immediately clear whether the secondary can grow sufficiently before its envelope has been ejected by the wind. Nevertheless, the primary white dwarf will at least be accreting from the secondary's wind and thus a system not unlike that of Mira (*o* Ceti) could form. In a more advanced phase of evolution with a higher mass loss rate from the secondary the system would look more and more like an extremely long-period symbiotic star. Eventually, with or without RLOF, the secondary will lose its hydrogen-rich envelope and become a white dwarf too.

We do not want to end this discussion without summarizing briefly the major simplifications and shortcomings in our numerical computations and commenting on their possible influence on the results obtained.

First, shortcomings are introduced by the stellar evolution code used to do the computations (for a brief description see Sect. 3). In the context of computing RLOF there are two simplifications which we wish to discuss briefly. The first is the use of a plane-parallel grey atmosphere in the Eddington approximation for determining the surface boundary conditions. While the assumption of a plane-parallel atmosphere can still be justified even in a case like ours where $H/R \approx 0.05$, the grey atmosphere approximation is not a particularly good one at low effective temperatures. However, since the surface conditions on the primary do not change drastically during the mass transfer phases computed, our results should at least be differentially independent of the atmosphere model used. The second simplification to be mentioned here is the assumption of thermal equilibrium i.e. $L_r = L =$ const in the outermost layers with a mass $0.97M = M_F \le M_r \le M$. Since the thermal time scale of such extended stars is very short (only a few years) and that of the surface layers with $M_e = M - M_F \approx 0.03M$ shorter still by about a factor M_e/M (see Eq. (14)), while the time scale of mass loss is of the order 10^5 yr, the assumption of thermal equilibrium in the envelope, though not correct, is probably of little consequence.

A second major shortcoming in our computations is that we did not perform a self-consistent binary evolution from the main sequence to RLOF. Rather, for reasons explained in Sect. 3, we bridged the gap between the first thermal pulse and the onset of mass transfer by using a semi-analytical description of the primary's evolution with wind mass loss on the AGB.

A third shortcoming is that we had to assume the value of the adiabatic mass radius exponent in order to constrain the parameters of systems that are dynamically stable against mass transfer. Therefore, we cannot be sure a priori that the systems we computed are really dynamically stable. However, the fact that we did not see any tendency of the mass transfer rate to run away (at least in the computations presented in Sect. 5) must mean that the growth time for a possible dynamical instability must be much longer than the time interval covered by our computations.

A fourth shortcoming concerns the mass loss via a stellar wind. Not only for simplicity but also because a substantially superior treatment is not available, we parametrized the wind loss rate according to the Reimers formula (Eq. (18)), although it is well known that the AGB evolution of single stars with only a Reimers-type wind does not reproduce well the observed initial to final mass relation of white dwarfs (see e.g. Weidemann and Koester, 1983). One of several possible reasons why the Reimers formula is not fully adequate is that stars terminate their AGB evolution by shedding mass at a much higher rate (via the so-called superwind) thereby producing the planetary nebula. In our computations we have neglected this possibility entirely and have forced the primary to lose its envelope via an ordinary Reimers wind or via RLOF. Because of our ignorance of the precise conditions for the ejection of a planetary nebula, we cannot be certain whether the evolution of the primary in our binary system would ever lead to the onset of RLOF. Nevertheless, the fact that we know at least one white dwarf, namely Sirius B, with a mass $M = 1.05\,M_{\odot}$ (Gatewood and Gatewood, 1978) demonstrates that there are stars whose AGB evolution ends at a much higher core mass than we envisaged here.

Another aspect that we have ignored completely in our computations is the possibility that the primary could pulsate as many stars high on the AGB are observed to do. Yet, even if we knew that the primary is pulsationally unstable, there is little we could have done, because treating RLOF from a pulsating donor star over time scales long compared to the pulsation period is beyond present computational capabilities.

8. Conclusions

The aim of this paper was to obtain some insight into the properties of binary systems in a phase of dynamically stable late case C mass transfer. With this end in view we explored in the first part of this paper (Sect. 2) the prerequisites for such an evolution to be possible, i.e. under which conditions the originally more massive primary can reduce its mass via a stellar wind on the AGB to the extent that the binary system will be dynamically stable against mass transfer at the onset of RLOF. We showed that such an evolution is in fact possible. Using the Reimers formula (Eq. 18) for the wind mass loss rate on the AGB, we provide quantitative estimates of the initial binary parameters that will result in this type of evolution (see Table 1).

In the second part (Sects. 3–6) we have studied the properties of binary systems with an initial primary mass of $\sim 3M_{\odot}$ during late case C mass transfer. As a first major result we demonstrate that numerical computations done with a standard technique for the computation of the mass transfer (presented in Sect. 3) give incorrect results. The failure of the standard technique can be traced back to the fact that the algorithm assumes implicitly a sharp stellar rim, whereas the photospheric scale height of the very extended primary ($R_1 \approx 500\,R_{\odot}$) is $H \approx 0.05 R_1 \approx 25\,R_{\odot}$.

An improved technique for computing mass transfer that takes the finite scale height of the stellar atmosphere into account is described in Sect. 4. Numerical simulations done with the improved technique (presented in Sect. 5) yield results that are qualitatively different from those obtained using the standard method. While the latter yields spasmodic mass transfer at each thermal pulse (with no mass transfer in between), the former yields sustained mass transfer with rates of the other 10^{-5}–$10^{-6}\,M_{\odot}\,\mathrm{yr}^{-1}$ modulated in time due to the occurrence of thermal pulses (see Figs. 8–10). Since the mass of the hydrogen-rich envelope of the primary is a few $0.1\,M_{\odot}$ and the mass loss rate of the order $10^{-6}\,M_{\odot}\,\mathrm{yr}^{-1}$ the mass transfer phase lasts a few 10^4–10^5 yr. Other important properties of the systems studied are summarized in Tables 3 and 4. Another remarkable property of binary systems of this kind is that the mass losing star is underfilling it critical Roche lobe significantly. In our examples: $R_1 \approx (0.7 - 0.8) R_{1,\mathrm{R}}$ (see Tables 3 and 4). This is entirely a consequence of the large photospheric scale height. (We note that in the standard terminology such systems would be called detached systems.) Furthermore, our computations in Sect. 5 show that the spurious dynamical mass transfer instability occurring with the standard method does not appear when the improved method is used unless the time scale on which the angular momentum that is transferred to the accretion disk is returned to the orbit exceeds 10^4 yr significantly.

Since the properties of the accretion disk that forms around the secondary may be decisive for the stability of mass transfer we explored them in some detail in Sect. 6. The main properties are the following. The disk is very large, its outer radius being $R_{\mathrm{D}} \approx 500 - 10^3\,R_{\odot}$. Its outer parts are very cool ($T_{\mathrm{D}} < 10^3$ K) and optically thick, the main source of opacity probably being dust. The vertical scale height is $H_{\mathrm{D}} \approx 0.1 R_{\mathrm{D}}$. Most important, its viscous time scale is very likely shorter than 10^4 yr. Therefore the disk cannot render the mass transfer dynamically unstable. Because the central object, i.e. the secondary, is a main sequence star, the disk's inner radius is of the order $\sim R_{\odot}$. As a consequence, the total accretion luminosity is always small compared to the luminosity of the primary. Therefore, such a disk will essentially remain invisible and manifest itself only by eclipsing the primary, given that the orbital inclination is high enough, i.e. $i \gtrsim 70°$.

The net result of this type of evolution will be a long-period binary system ($P \gtrsim 10$ yr) consisting of a white dwarf and a main sequence star with a mass of a few solar masses.

Acknowledgements. The authors are grateful to Dr. G. Hensler, F. Meyer, E. Meyer-Hofmeister, H.-C. Thomas and A. Weiss for helpful discussions and to Dr. K. Butler for improving the language of the manuscript. This work was supported by the Deutsche Forschungsgemeinschaft, grants Ku 474/8-2 and Ku 474/13-1.

References

Alexander, D.R.: 1975, *Astrophys. J. Suppl.* **29**, 363
de Kool, M., van den Heuvel, E.P.J., Rappaport, S.A.: 1986, *Astron. Astrophys.* **164**, 73
D'Antona, F.: 1982, *Astron. Astrophys.* **114**, 289
Duschl, W.J.: 1986, *Astron. Astrophys.* **163**, 56
Eggleton, P.P.: 1983, *Astrophys. J.* **268**, 368
Flannery, B.P.: 1975, *Monthly Notices Roy. Astron. Soc.* **170**, 325
Frank, J., King. A.R., Raine, D.J.: 1985, *Accretion power in astrophysics*, Cambridge, Cambridge University Press
Gatewood, G.D., Gatewood, C.V.: 1978, *Astrophys. J.* **255**, 191
Hensler, G.: 1982, *Astron. Astrophys.* **114**, 319
Hjellming, M.S., Webbink, R.F.: 1987, *Astrophys. J.* **318**, 794
Hofmeister, E., Kippenhahn, R., Weigert, A.: 1964, *Z. Astrophys.* **59**, 215
Iben, I. (Jr.): 1986, *Astrophys. J.* **304**, 201
Kettner, K.U., Becker, H.W., Buchmann, L., Görres, J., Kräwinkel, H., Rolfs, C., Schmalbroch, P., Trautvetter, H.P., Vlicks, A.: 1982, *Z. Phys. A* **308**, 73
Kippenhahn, R.: 1981, *Astron. Astrophys.* **102**, 293
Kippenhahn, R., Weigert, A., Hofmeister, E.: 1967, *Methods for Calculating Stellar Evolution*, in *Methods in Computational Physics*, eds. B. Alder, S. Fernbach, M. Rotenberg, New York, Academic Press, Vol. 7, p. 129

Kippenhahn, R., Weigert, A.: 1967, *Z. Astrophys.* **65**, 251
Kippenhahn, R., Meyer-Hofmeister, E.: 1977, *Astron. Astrophys.* **54**, 539
Lauterborn, D.: 1970, *Astron. Astrophys.* **7**, 150
Lubow, S.H., Shu, F.H.: 1975, *Astrophys. J.* **198**, 383
Mazzitelli, I., D'Antona, F.: 1986, *Astrophys. J.* **311**, 762
Meyer, F.: 1986, in *Radiation Hydrodynamics in Stars and Compact Objects*, eds. D. Mihalas, K.-H. A. Winkler, *Lect. Notes Phys.* **255**, Springer, Berlin, Heidelberg, New York, p. 249
Meyer, F., Meyer-Hofmeister, E.: 1983, *Astron. Astrophys.* **121**, 29
Mochnacki, S.W.: 1984, *Astrophys. J. Suppl. Ser.* **55**, 551
Neo, S., Miyaji, S., Nomoto, K., Sugimoto, D.: 1977, *Publ. Astron. Soc. Japan* **29**, 249
Packet, W.: 1981, *Astron. Astrophys.* **102**, 17
Paczyński, B.: 1966, *Acta Astron.* **16**, 231
Paczyński, B.: 1970, *Acta Astron.* **20**, 47
Paczyński, B.: 1975, *Astrophys. J.* **202**, 558
Paczyński, B., Sienkiewicz, R.: 1972, *Acta Astron.* **22**, 73
Plavec, M., Kratochvil, P.: 1964, *Bull. Astron. Inst. Czech.* **15**, 165
Plavec, M., Ulrich, R.K., Polidan, S.: 1973, *Publ. Astron. Soc. Pacific* **85**, 769
Pringle, J.E.: 1985, in *Interacting Binary Stars*, eds. J.E. Pringle, R.A. Wade, Cambridge, Cambridge University Press, p. 1
Reimers, D.: 1975, in *Problems in Stellar Atmospheres and Envelopes*, eds. B. Baschek, W.H. Kegel, G. Traving, Springer, Berlin, Heidelberg, New York, p. 229
Ritter, H.: 1988, *Astron. Astrophys.* **202**, 93
Savonije, G.J.: 1978, *Astron. Astrophys.* **62**, 317
Shakura, N.I., Sunyaev, R.A.: 1973, *Astron. Astrophys.* **24**, 337
Smak, J.: 1984, *Publ. Astron. Soc. Pacific* **96**, 5
Taam, R.E., Mc Dermott, P.N.: 1987, *Astrophys. J. Letters* **319**, L83
Thomas, H. -C.: 1977, *Ann. Rev. Astron. Astrophys.* **15**, 127
Webbink, R.F.: 1979a, in *Changing Trends in Variable Star Research*, IAU Coll. No. **46**, eds. F.M. Bateson, J. Smak, I.H. Urch, Hamilton, New Zealand, University of Waikato Press, p. 102
Webbink, R.F.: 1979b, in *White Dwarfs and Variable Degenerate Stars*, IAU Coll. No. **53**, eds. H.M. van Horn, V. Weidemann, Rochester, University of Rochester, p. 426
Webbink, R.F.: 1985, in *Interacting Binary Stars*, eds. J.E. Pringle, R.A. Wade, Cambridge, Cambridge University Press, p. 39
Webbink, R.F., Rappaport, S., Savonije, G.J.: 1983, *Astrophys. J.* **270**, 678
Weidemann, V., Koester, D.: 1983, *Astron. Astrophys.* **121**, 77
Weiss, A.: 1986, Dissertation, Ludwig-Maximilians-University, Munich
Weiss, A.: 1987, *Astron. Astrophys.* **185**, 165